

Research paper

Empirical design and comprehensive analysis of multiport converter in distributed generation enabled microgrids

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ABSTRACT

Distributed generation-enabled microgrids are a revolutionary approach for the energy paradigms that will transform the ways of generation, distribution, and utilization. A decentralized approach to energy generation in microgrids results in the robustness and independence of energy networks. Empowerment of the communities to take control of energy requirements with multiport converter-based microgrids promotes sustainability, efficiency, reduced losses, and increased resilience against the evolving trends of the energy sector. In this research, a hybrid energy-integrated multiport converter is designed and analyzed under its various modes of operation. Rapid transitions from one energy source to another are attained within 0.4 μs to feed the load at a steady state voltage of 400 V, and within 0.15 μs to charge the energy storage system at a steady state voltage of 48 V which indicates the robustness and independence of energy systems via this proposed approach. The resilience of the microgrid is obtained because of this rapid transition. Moreover, the proposed converter offers a voltage gain of 8.3 with 96 % efficiency making it sustainable for maximization of renewable energy. In addition, the resilience of the proposed converter is exhibited through high power densities at power losses of not more than 7.5 % making it an innovative candidate for hybrid energy systems as compared to existing technologies. Simulation scenarios are validated through mathematical analysis, proving their effectiveness in distributed generation-enabled microgrids. The proposed research model is effective in a real environment because of the minimal effects of parasitic resistance, reduced total harmonic distortion under different loads, and negligible electromagnetic interferences, demonstrated through simulations.

1. Introduction

An energy source derived from sustainable natural resources is not endless when used. To avoid the rising temperature of the climate, renewable energy plays an important role. This is because it does not produce harmful gases such as nitrogen, carbon dioxide, and many other greenhouse gases, that hand out to global warming (Moiceanu and Dinca, 2021), (Reid and Goldemberg, 1998), (Tavassoli and Kamran-Pirzaman, 2023), (Wang et al., 2023), (Liu and Han, 2024), (Li et al., 2023), (Justice et al., 2024).

For a century, the demand for electricity for industries, residents, and vehicles has been heavily dependent on the energy from fossil fuels such as coal, oil, etc. In the future, fossil fuels will be in shortage and their

prices will be high. Additionally, fossil fuel energy is not friendly to the environment, producing harmful greenhouse gases (Di Filippo et al., 2022), (Kabir et al., 2023). The ozone layer protects the environment from the sun's harmful gases and maintains the earth's temperature. The greenhouse gases damage the ozone. As a result, climate changes are inevitable. Coal, oil, and gas are the major stakeholders of global climate change. Over 75 percent of global greenhouse gas emissions and nearly 90 percent of all carbon dioxide emissions are due to fossil fuels (U. Nations, 2024). Climate changes raise the sea level (Mimura, 2013) and affect wildlife habits (Schmitz and Sylvén, 2023), (Abrahms et al., 2023). It is estimated from Harvard's calculation that air pollution from fossil fuels may be responsible for as many as one in five human deaths worldwide (Renewable Energy's Benefits for Air Quality and Climate,

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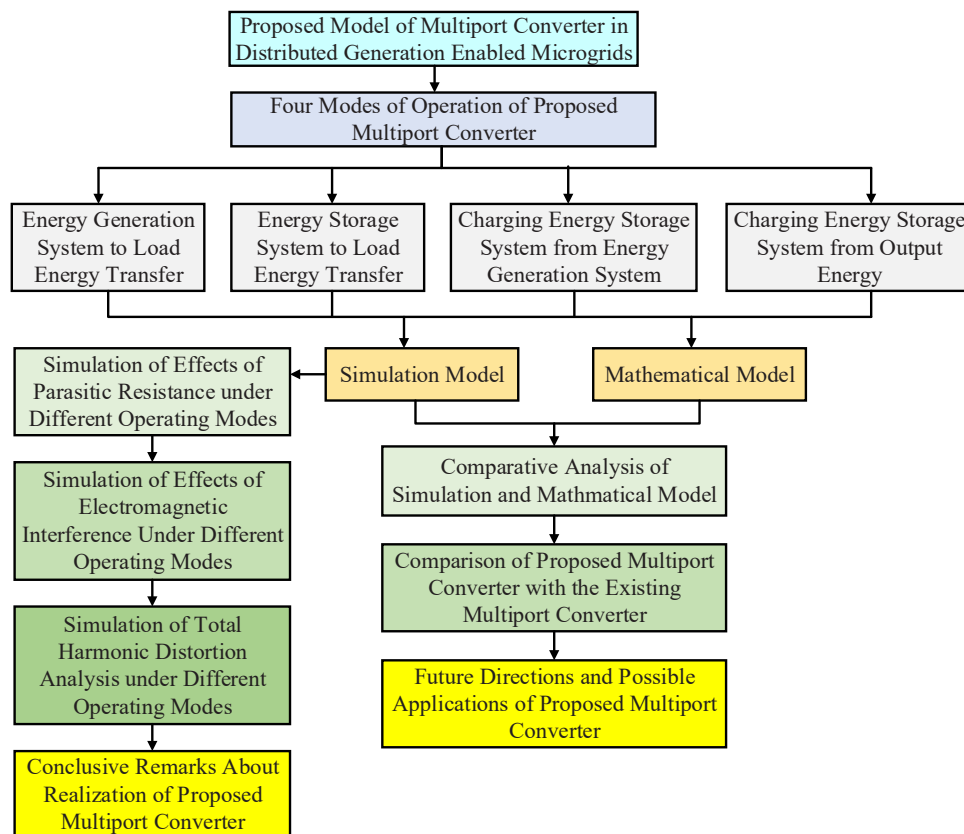


Fig. 1. Different stages of research carried out from the beginning to the end.

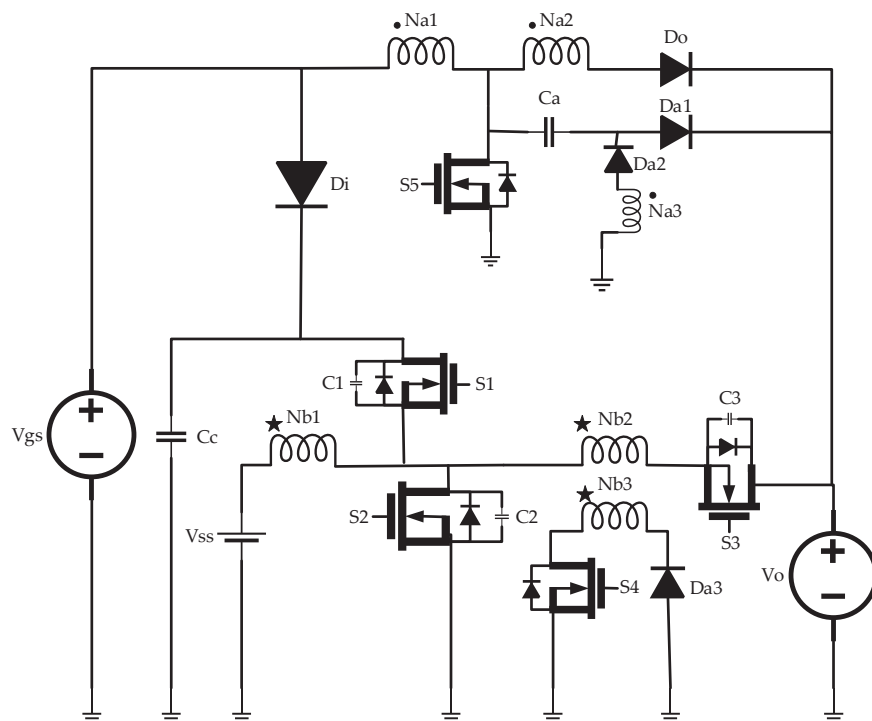


Fig. 2. Proposed circuit of multi-port DC-DC converter based on soft-switching mechanism.

2024).

Clean energy nowadays provides different opportunities such as the source of new jobs, increases the flexibility of electrical grids, easy electrical supply in the country, and most importantly reduces the tariff

(Ang et al., 2022), (Owusu and Asumadu-Sarkodie, 2016), (Seminario-Córdova and Rojas-Ortega, 2023).

Being a responsible community of the universe, it is important to explore new energy sources and ways to resolve existing problems of

Table 1
DIFFERENT PARAMETERS OF PROPOSED CIRCUIT OF MULTI-PORT DC-DC CONVERTER.

Sr. No.	Circuit Parameter	Explanation
1	V_{gs}	Energy generation system voltage
2	V_{ss}	Energy storage system voltage
3	V_o	Output voltage
4	$N_{a1}, N_{a2}, \text{ and } N_{a3}$	Coupled inductors connected to the energy generation system
5	$N_{b1}, N_{b2}, \text{ and } N_{b3}$	Coupled inductors connected to the energy storage system
6	$C_1 \text{ and } C_2$	Snubber capacitors
7	$S_1, S_2, S_3, S_4, \text{ and } S_5$	Power electronic switches
8	C_a	Lossless passive snubber capacitor
9	D_o	Boost converter output diode
10	$D_{a1}, \text{ and } D_{a2}$	Lossless passive snubber diodes
11	L_{M1}	Boost converter inductor
12	L_{M2}	Buck converter inductor
13	$L_{LK1}, L_{LK2}, \text{ and } L_{LK3}$	Leakage inductance

energy shortages and carbon emissions. Solar and wind are an important source of renewable and sustainable energy. There is a rapid incline of human interest toward the adaptation of solar and wind-based renewable energy in the second decade of the 21st century. Approximately 65 percent of the world's total electricity supply can be availed from economical renewable energy sources by 2030. Decarbonization of the power sector can be attained by 2050 through the adaptation of renewable energy. This will reduce carbon emissions and help to mitigate climate change (Renewable Energy's Benefits for Air Quality and Climate, 2024).

Behind so many advantages, some challenges are associated with solar and wind-based renewable energy. Their intermittent nature creates grid stability issues (Hossain et al., 2023) (Khalid, 2024). The outdated transmission and distribution system is reluctant to integrate variable renewable energy sources (Saha et al., 2023). Expensive and non-scalable energy storage systems result in loss of energy and unreliability (Sayed et al., 2023). Therefore, continuous innovation and

investment are required to overcome such obstacles and achieve sustainable energy in the future.

The continuous research introduces the concept of hybrid energy systems. Hybrid energy systems are comprised of conventional, renewable, and storage systems. The integration is made possible between different energy systems with the aid of multiport converters. Multiport converters are advanced power electronic converters that are capable of handling multiple inputs and outputs simultaneously. Hence, different generation sources like wind, solar, and storage batteries can be controlled efficiently for flexible energy management. This capability gives rise to distributed generation-enabled microgrids.

The main objective of this research is to propose and analyze the technique for the maximization of renewable energy sources and minimization of carbon emissions-based energy sources. In this research, a hybrid energy system's integrated multiport converter is developed and analyzed. A hybrid energy system combines several energy-generating sources and energy storage systems. The usage of hybrid electric cars and DC micro-grid applications is expanding, which hastens the development of hybrid energy systems. Some DC-DC converters are used in these systems to control the energy exchange between the load and various input sources (Saritha et al., 2023), (Faraji et al., 2020), (Khosrogorji et al., 2016), (Farajadian et al., 2024), (Faraji and Farzanehfard, 2018). Multi-port converters have emerged as a result of the design tendency of combining converters to boost power density. In other words, an integrated multi-port converter may be employed with fewer parts than numerous independent converters with their specific input sources. The use of these converters in hybrid energy applications is also expanded by the presence of energy storage systems and their charge ability from the output in multi-port converters (Gevorkov et al., 2023), (Madhana and Mani, 2022a), (Madhana and Mani, 2022b).

Providing multiple power flow paths between the inputs and the output, independence of the power flow paths to have better power management, integrating and sharing components to reduce the component count, using high step-up techniques to enhance the voltage gain, and providing soft switching conditions to increase efficiency are some of the challenges that must be taken into consideration during the design process of multi-port converters.

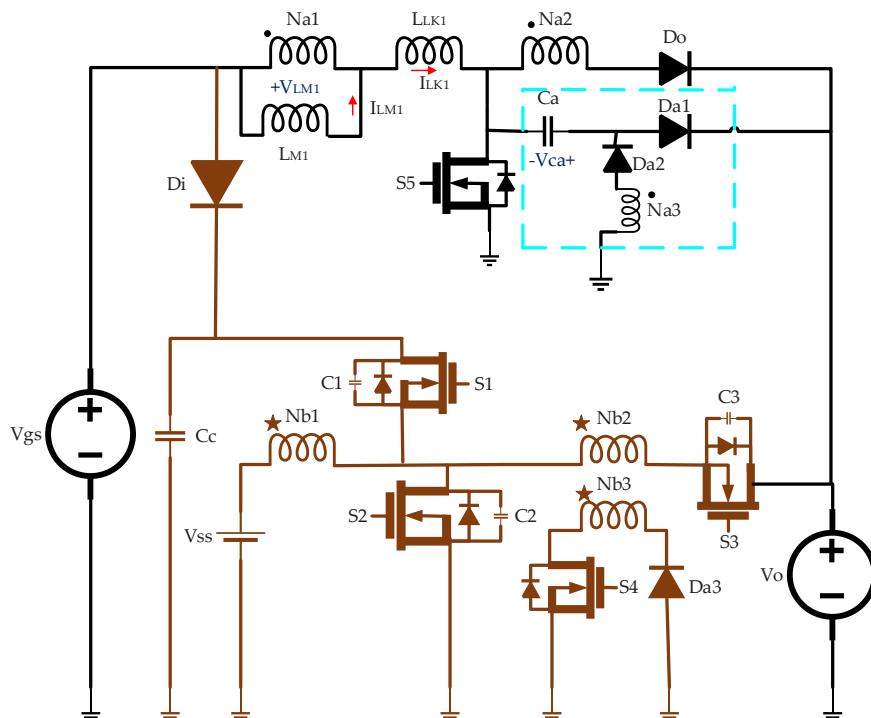


Fig. 3. The energy transfer from energy generation system to load.

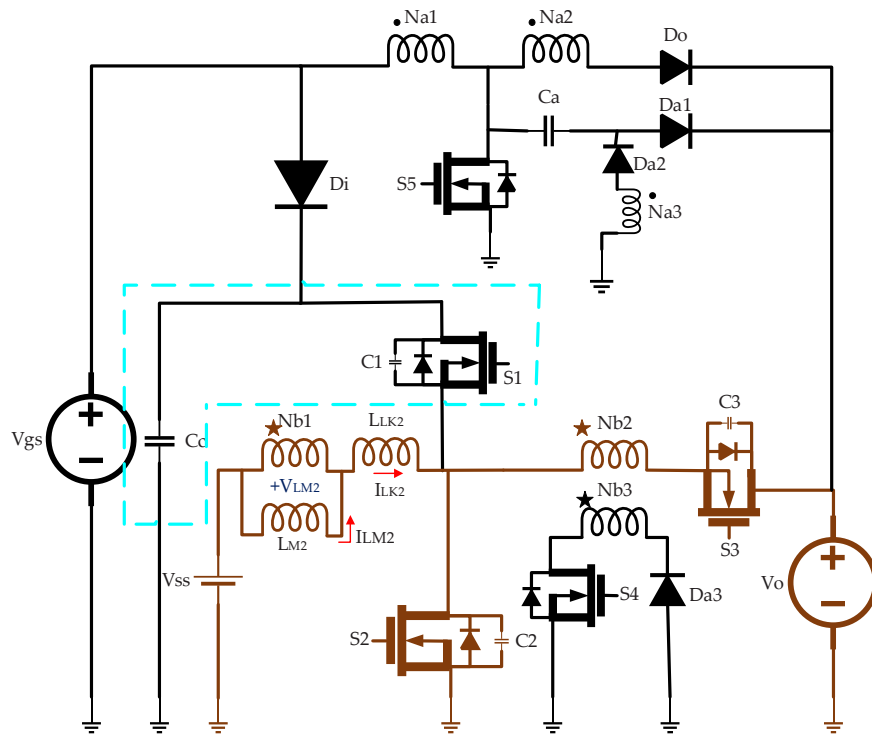


Fig. 4. The energy transfer from energy storage system to load.

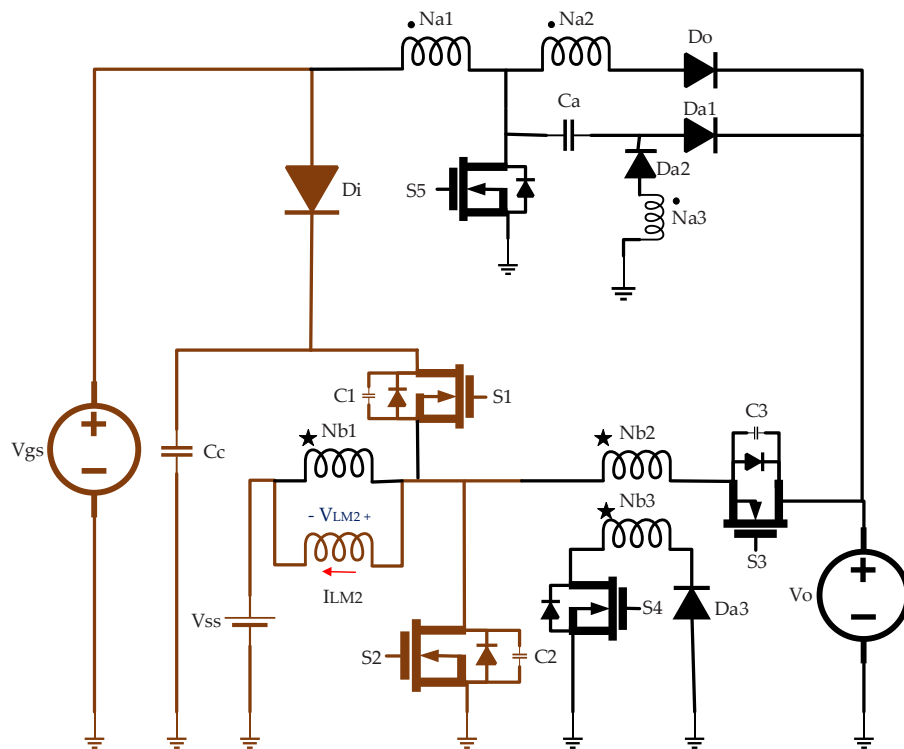


Fig. 5. The charging of the energy storage system from the energy generation system.

It is crucial that energy storage systems can be charged simultaneously from both energy generating systems and the output DC link for modern applications like the smart grid and electrical hybrid autos.

The literature describes a variety of non-isolated multi-port DC-DC converters to overcome some of the challenges stated above.

2. Literature review

In the past, a multi-input converter was created by concurrently connecting many DC/DC converters (Zhiyun Qian et al., 2010), (Shen and Yang, 2013), (Tao et al., 2006), (Mei et al., 2008), (Yu and Kwassinski, 2013), (Zeng et al., 2014). The use of these methods causes

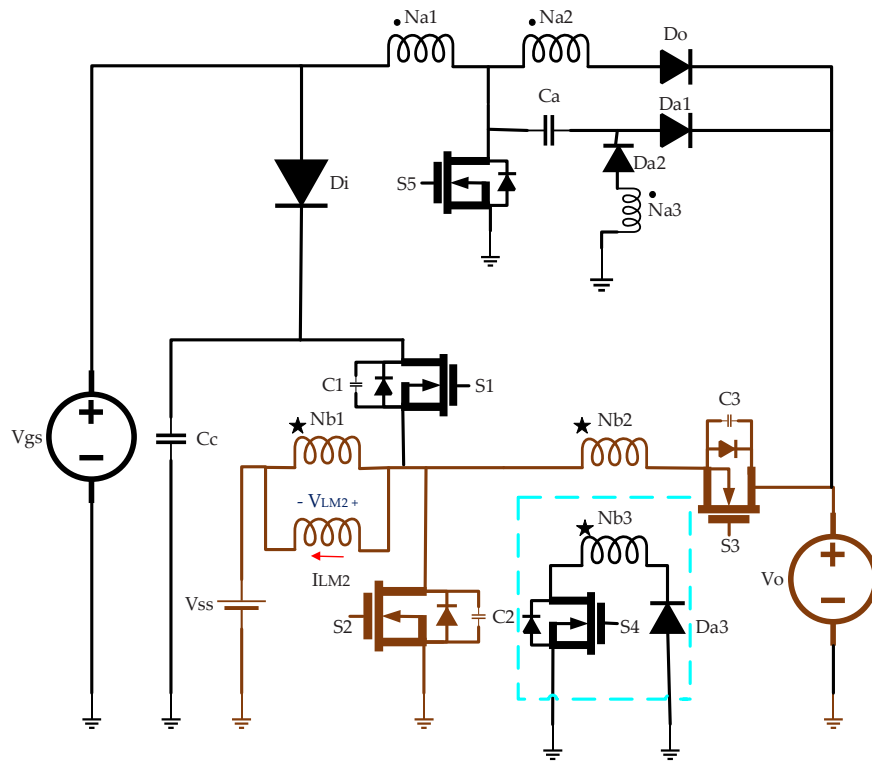


Fig. 6. The charging of the energy storage system from the output.

several issues, such as voltage fluctuation and protection issues. The three fundamental categories are electrical, magnetic, and electromagnetic. The first group of topologies contains the multi-winding transformer. They are referred to as multi-input converters (Chen et al., 2002). The second group configurations have a DC link. Both the final source and the sources that are attached to it have a two-winding transformer and a DC link (Tao et al., 2005), (Harbi et al., 2023). Many input devices may be divided primarily into three categories. Three via electrical-magnetically coupling, two by electrical coupling, and one by magnetic coupling. The characteristics used to categorize converters include battery life cycles, gentle switching, source operation, isolation between output and input, and more. All of the aforementioned characteristics are shared by joint DC-Link converters (Tao et al., 2005), magnetically coupled converters (Repecho et al., 2021), and boost-integrated phase-shift full-bridge converters (Al-Atrash and Batarseh, 2007).

The number of inputs is increased, and an energy storage system charging method is provided, using a straightforward diode-switch architecture (Faraji et al., 2019), (Chien et al., 2014). Sharing the power flow channel among the input sources reduces the number of components, but adding more inputs limits the energy contribution of each source. Additionally, since not all switches are soft switches, charging the energy storage device from the output is not practical.

By combining fundamental converters, an inductance-based multi-port converter is produced (Cheng et al., 2018). Using a single inductor for all working modes reduces the converter's size, but it prohibits the modes from having separate power flow routes. The switches' ability to switch hard has an impact on the converter's effectiveness. Two boost converters with linked inductors are connected in series to create a high step-up multi-port converter (Chen et al., 2013). The use of two active clamp circuits would be necessary to provide a soft switching state. A lower component count is achieved by sharing the power flow channel between two converters; however, the power management of each input source depends on the state of the other input source. Even though just one of the inputs supplies the load, this converter has five switches that

must be flipped for it to operate in any mode. Conduction loss is thus a factor in converter efficiency (Jaschke, 2007).

Using different converters without sharing their components with other converters is one way to create multi-port converters that can work independently from one another (Wai and Chen, 2014), (Monteiro et al., 2018). A bidirectional power flow line may be used to link two different converters to two distinct input sources, and the output DC connection can be utilized to charge an energy storage device. The converter operates in boundary conduction mode, which provides gentle switching conditions. Both converters do not experience a significant voltage gain since the switches do not function under soft-switching conditions.

Another technique for creating different power flow paths from inputs to outputs is modular architectures (Hintz et al., 2015), (Varesi et al., 2019). With this strategy, each input receives a full converter, and the converters' output is simply shared. A multi-port buck-boost DC-DC converter for use in hybrid electric vehicle applications is created by using a modular design (Hintz et al., 2015). This converter offers several power pathways between the inputs and the output while providing seamless switching. The quantity of inductors is also decreased by using shared inductors in various operating modes. The converter's primary drawbacks are the use of six switches and the low voltage gain. To create a multi-port converter, a boost-based or bidirectional converter may be added as a new module (Varesi et al., 2019). It relies on either the energy generating system or the energy storage system. Despite operating under challenging switching circumstances, this converter offers high gain voltage. However, the complex structure of this method is still a concern.

The energy-producing system and the output of the unique soft switched non-isolated high step-up multi-port DC-DC converter proposed in this study may be used to charge the energy storage system. In addition to having a low component count and a small number of components, the converter also has a high step-up voltage gain and gentle switching circumstances. To provide gentle switching circumstances, four separate procedures are successfully used without

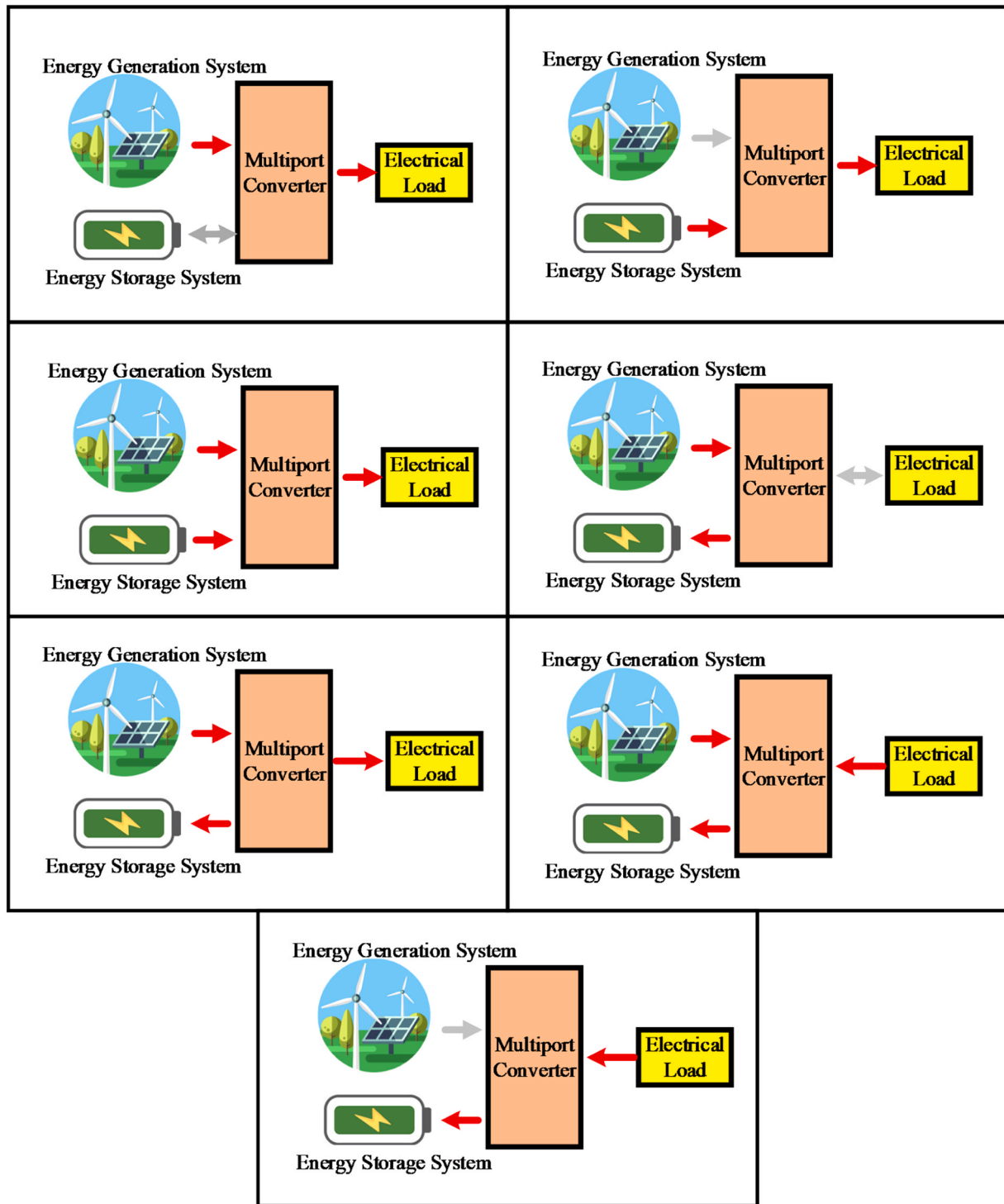


Fig. 7. The different operating conditions of the proposed multiport converter.

interfering with one another, and each input source has its dedicated power flow line. As a result, in all operating modes, all switches operate in the soft switching condition. By exchanging components between modes and using an energy storage system with a bidirectional path, the number of converter components is also reduced. Power conversion is finished in a single step in all operational modes, which lowers conduction loss. The final result is an efficient and well-integrated multiport converter. Moreover, high voltage gain and reduced current ripples make the proposed multiport converter ideal for variable renewable energy sources. The proposed model offers reduced total harmonic

distortion under different operating load conditions. This feature makes it an ideal candidate for distributed loads and generations. Further, the snubber circuits used in the proposed research minimize the electromagnetic interferences which in turn eliminates the filtering requirements. A flow chart is added in Fig. 1 that involves the different stages of research.

This research paper is divided into nine sections. The construction and operational modes of the suggested converter are described in Section III. In section IV, the mathematical analysis of the converter and its design issues are discussed. Simulation results and discussion are

Table 2

Values of different parameters of microgrid's test converter.

Sr. No.	Parameters	Values
1	L_1	577 μ H
2	L_2	2692 μ H
3	L_3	5250 μ H
4	C_a	5nF
5	C_1	1 μ F
6	C_2	5 μ F
7	V_1	96 V
8	V_2	48 V
9	S_1, S_2, S_3	IRFP260
10	V_{out}	400 V
11	D_1, D_2, D_3	RHRP15120

Table 3

Different operating cases of the test system.

Cases	Description
Case I	Energy Generation System to Load Energy Transfer
Case II	Energy Storage System to Load Energy Transfer
Case III	Energy Storage System Charging by Energy Generation System
Case IV	Energy Storage System Charging from Output Energy

added in Section V. The validation of the simulation results by mathematical calculations and their significance is presented in Section VI. The performance comparison of the proposed multiport converter for microgrid application with the existing multiport converters is added in section VII. The future applications of the proposed converter are explained in Section VIII. The conclusions are presented in Section IX.

3. Proposed operational design of multiport converter

The suggested converter's construction and the equivalent circuit for each operating mode are shown in Fig. 2. The converter that is being suggested has two stages. In the upper phase, V_{gs} can be connected to an energy-generating device, such as fuel cells or photovoltaic (PV) cells, and transmit energy to an output DC connection. An energy storage device that can be directly charged by V_{gs} and has a bidirectional power

flow line with the output, is linked to V_{ss} in the lower phase. The parameters of the proposed circuit of the multi-port DC-DC converter are presented in Table 1.

The advantages of the proposed multiport converter are not limited to the integration of multiple renewable energy sources for the establishment of microgrids, and enhanced energy management with increased efficiency and reduced components, but also offers redundancy, reliability of power supply, and plugin of electric vehicles.

The proposed converter has four alternative working modes that depend on the charge status of V_{ss} , the amount of energy generated from V_{gs} , and the load power requirement. These are:

- Energy Generation System to Load Energy Transfer
- Energy Storage System to Load Energy Transfer
- Energy Storage System Charging by Energy Generation System
- Energy Storage System Charging by Output Energy

3.1. Energy generation system to load energy transfer

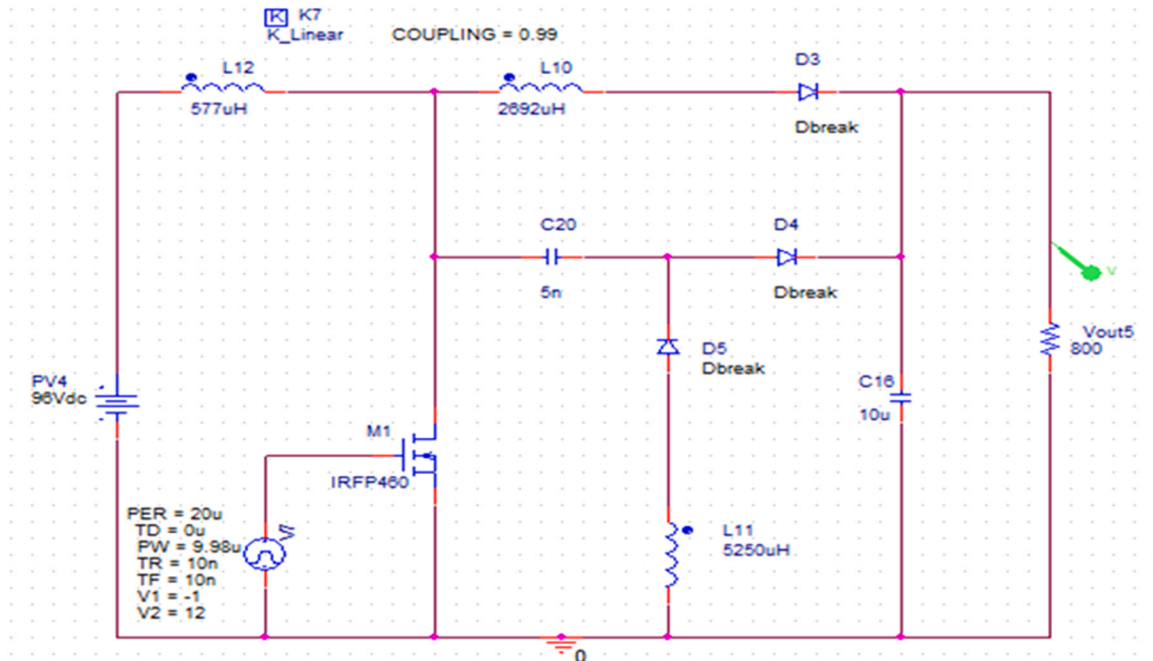
Fig. 3 presents the elements that go into creating this operating mode. This converter's part, which functions separately from the other converter sections, directly transmits energy from the energy generation system to the load.

To increase the voltage gain in this working mode, a boost converter with connected inductors ($N_{a1} - N_{a2}$) is required. Additionally, D_o is the boost converter output diode, and S_5 is the boost converter switch. A passive lossless snubber circuit made of C_a , D_{a1} , D_{a2} , and N_{a3} is employed to offer S_5 with zero voltage switching turn-off (Renewable Energy's Benefits for Air Quality and Climate, 2024).

Together with N_{a1} and N_{a2} , N_{a3} is wound around a single magnetic core. Additionally, the N_{a1} and N_{a2} windings are connected to S_5 , enabling zero current switching.

3.2. Energy storage system to load energy transfer

The components that work in this operating mode are shown in Fig. 4. In this section of the proposed converter, a boost converter with coupled inductors ($N_{b1} - N_{b2}$) is used to transport energy from the

**Fig. 8.** Simulation circuit of transferring energy from the energy generation system to the load.

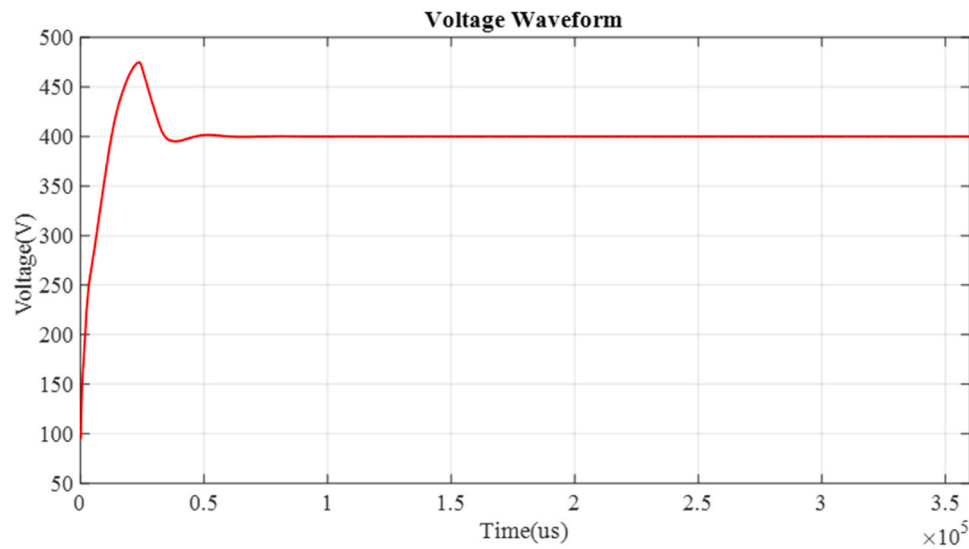


Fig. 9. Voltage waveform of transferring energy from the energy generation system to the load.

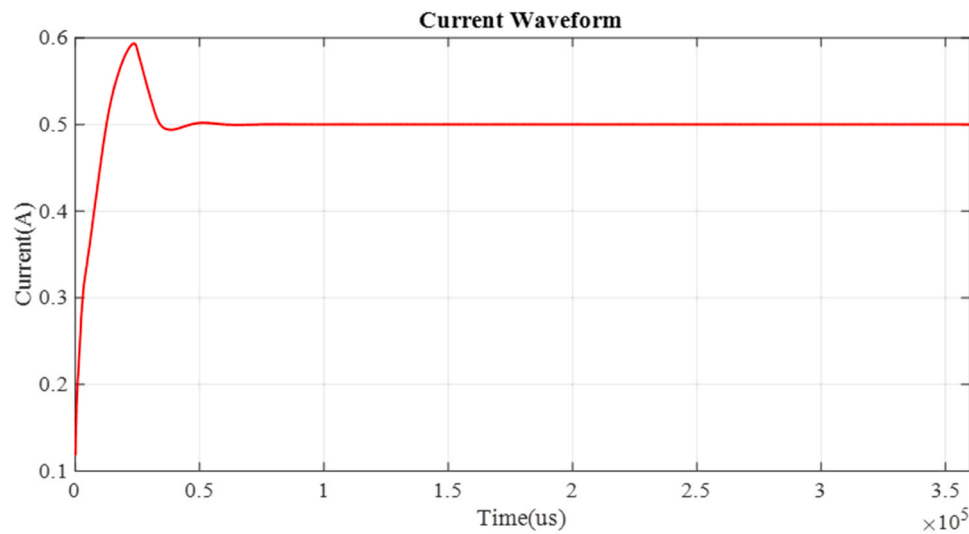


Fig. 10. Current waveform of transferring energy from the energy generation system to the load.

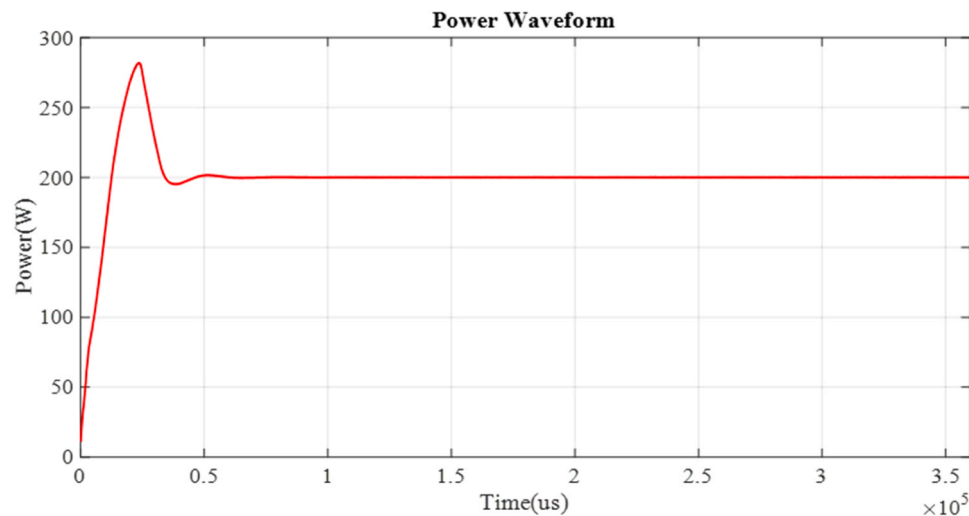


Fig. 11. Power waveform of transferring energy from the energy generation system to the load.

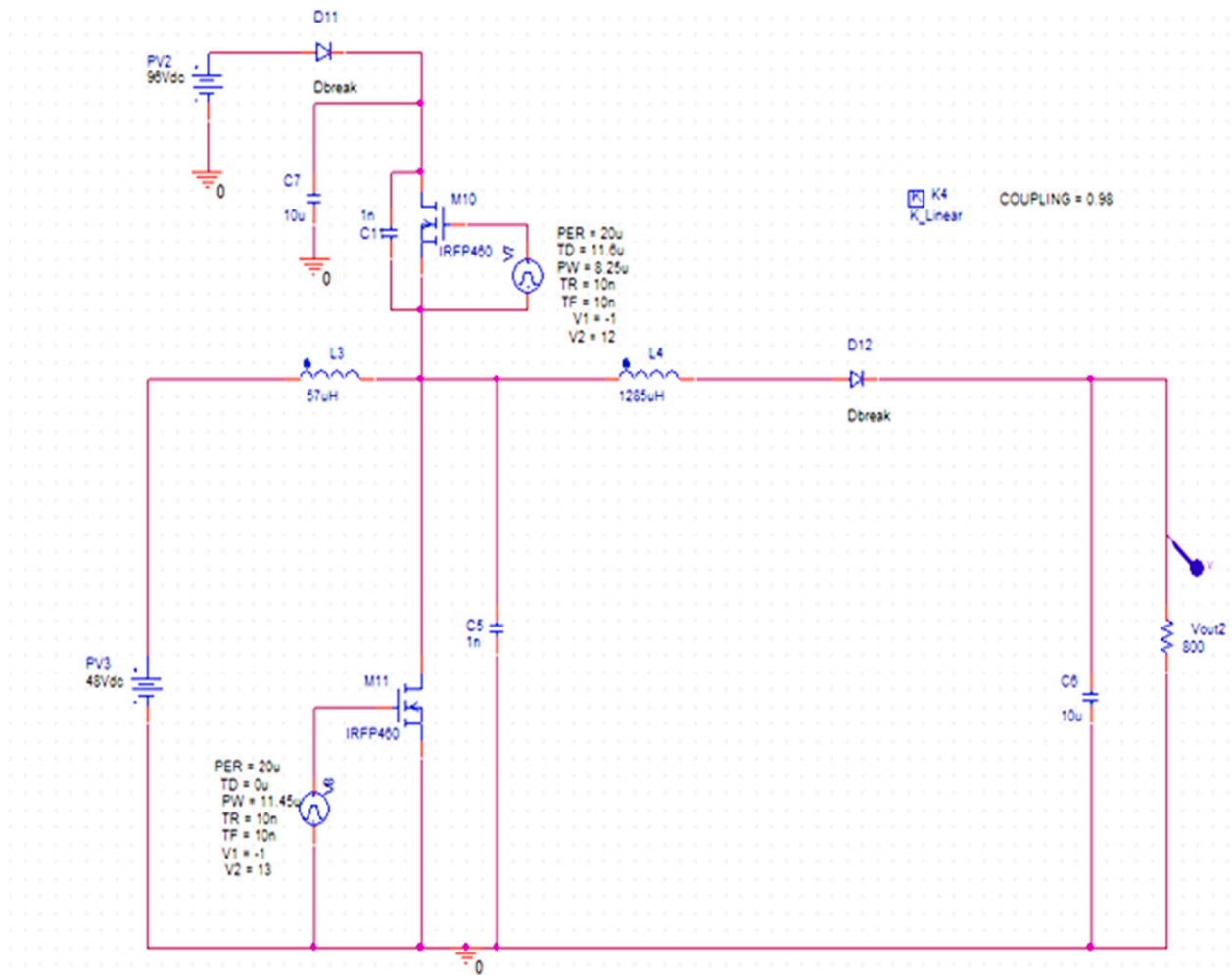


Fig. 12. Simulation circuit of transferring energy from the energy storage system to the load.

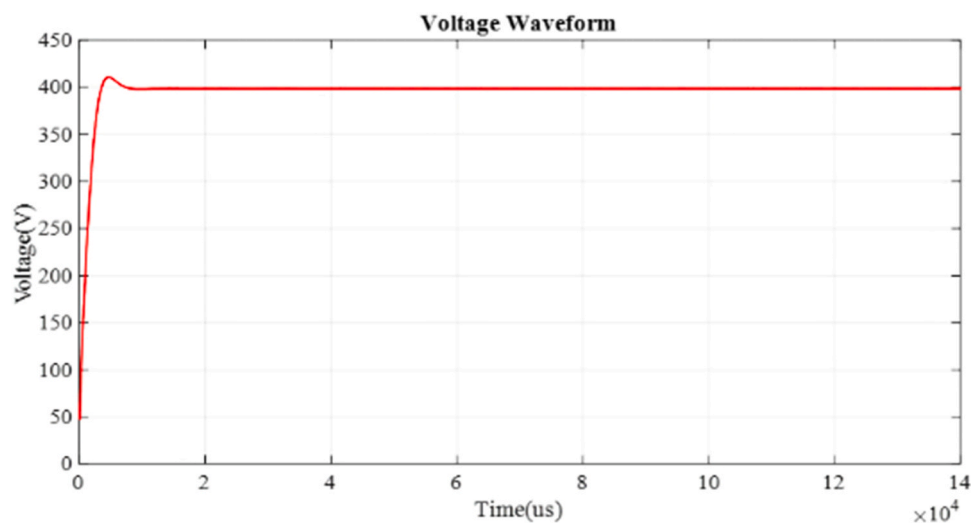


Fig. 13. Voltage waveform of transferring energy from the energy storage system to the load.

energy storage system to the load. The body diode S_3 serves as the boost converter diode, while S_2 acts as the converter's main switch. To create a soft switching environment, an active clamp circuit using S_1 and C_c is employed. The zero-voltage switching-based turning-off conditions are attained by the snubber capacitors C_1 and C_2 for S_1 and S_2 , respectively.

Leakage inductance discharges the capacitors' stored energy through the active clamp circuit before the switches are turned on.

As a result, under the zero-voltage switching environment, the S_1 and S_2 are switched. V_{cc} is significantly greater than V_{gs} in this operating state, and D_{bc} is reverse-biased, separating the upper side from the lower

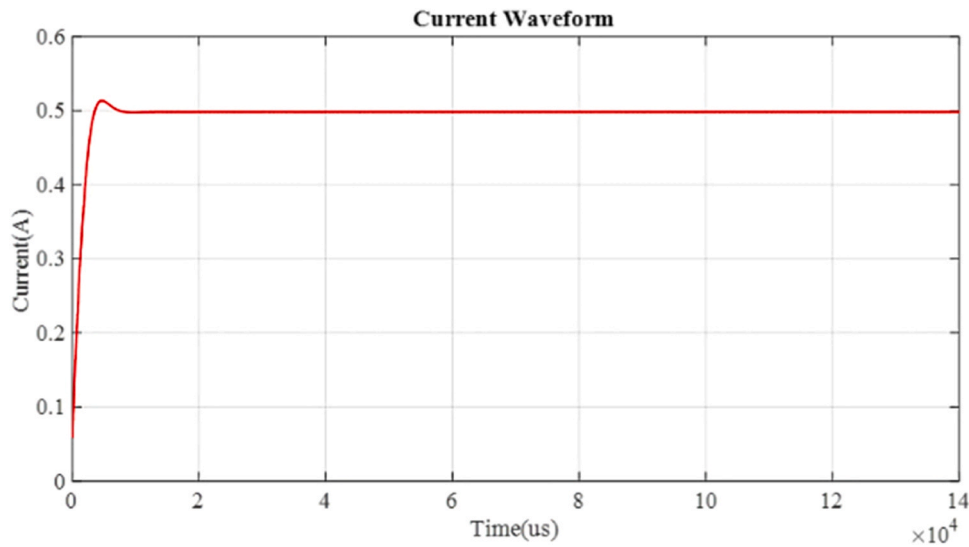


Fig. 14. Current waveform of transferring energy from the energy storage system to the load.

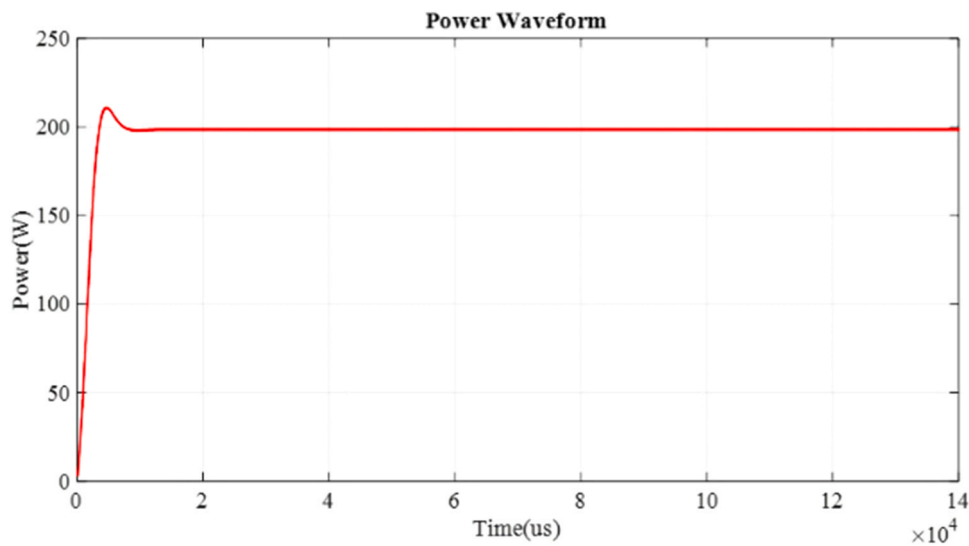


Fig. 15. Power waveform of transferring energy from the energy storage system to the load.

side of the converter.

3.3. Energy storage system charging by energy generation system

In this operating environment, the synchronous buck converter as shown in Fig. 5, consists of S_1 as the main switch, S_2 as the synchronous rectifier, and L_{M2} as the buck converter inductor would be used by the energy generation system to charge the energy storage system, resulting in a change in behaviors of several lower phase components.

Because the S_3 body diode is reverse biased, there is no current flowing through the main and secondary sides of the coupled inductors during this operating mode.

In this functioning state, the zero-voltage switching turning-off condition is established by capacitors C_1 and C_2 . These capacitors act as snubber capacitors. The boundary conduction mode of the converter is established to accomplish the zero-voltage switching based on turning on condition by taking into consideration the design of the L_{M2} value (Baek et al., 2013). The L_{M2} value should be within boundary conduction mode.

3.4. Energy storage system charging from output energy

The components in the lower phase of the proposed converter alter their behavior and function as a bidirectional converter when the output energy charges an energy storage system, as shown in Fig. 6. S_3 is the primary switch and S_2 is the buck converter's synchronous rectifier in this converter. Inductors ($N_{b1} - N_{b2}$) that are linked together are utilized to lower the output voltage. Additionally, a zero-voltage-transition circuit made up of S_4 , N_{b3} , L_{LK3} , and D_{a3} components provide a soft switching environment for S_2 , S_3 , and S_4 Farajdadian et al., (2024). Conventionally, the auxiliary switch (S_4) in zero-voltage transition circuits switches under a zero-current switching environment and provides a zero-voltage switching environment for the converter's other switches (S_2 and S_3).

There are different working modes in the multiport DC-DC converter, which may charge an energy storage system from an energy generating system and the output, as illustrated in Fig. 7. Other operating circumstances include a mix of the aforementioned operating modes since each portion of the proposed converter can function independently.

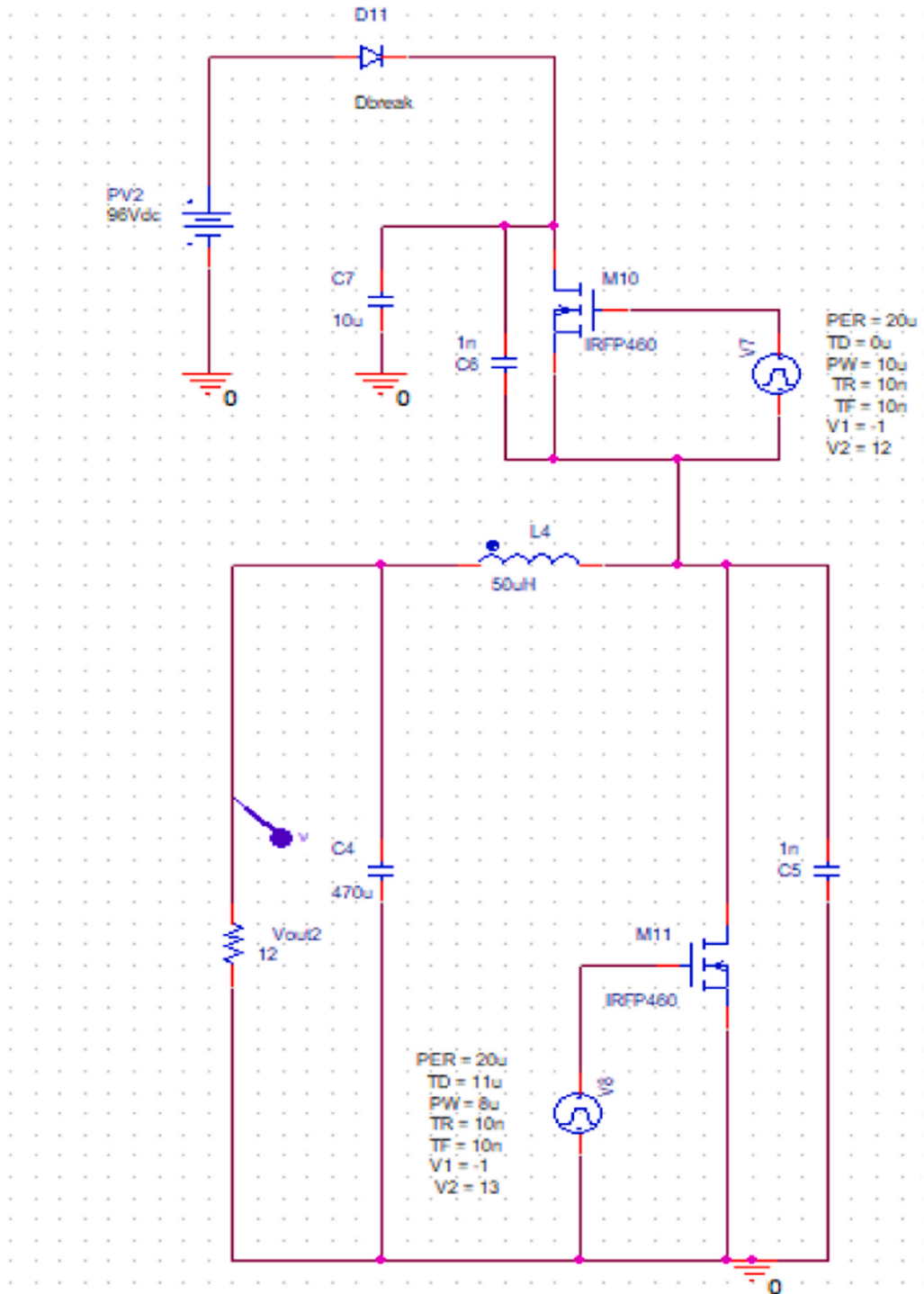


Fig. 16. Simulation circuit of charging energy storage system from energy generation system.

4. Mathematical analysis of proposed converter

The real challenges faced in the converter's operational design are integration, shared mechanism of components, and provision of soft switching in different operating conditions. Initially, the operating condition of each design of the converter is required to be analyzed.

Based on the analysis, a mechanism is then formulated for trading off among parameters associated with the different operating conditions of the converter. Some assumptions aid in simplification of the analysis, considering all circuit elements ideal. The voltages across snubber capacitance C_a and clamping capacitance C_c are considered constant.

The steady-state condition is assumed for the operation of the converter.

The turn ratios of the upper (a-phase) and lower (b-phase) phases of the converter are given as $n_1 = N_{a2}/N_{a1}$, $n_2 = N_{a3}/N_{a1}$, $n_3 = N_{a3}/N_{a2}$, $n_4 = N_{b2}/N_{b1}$, $n_5 = N_{b3}/N_{b1}$ and $n_6 = N_{b3}/N_{b2}$ respectively.

In the operating condition of the energy generation system to load energy transfer, the voltage gain is expressed as:

$$\frac{V_{out}}{V_{in}} = \frac{1 + n_1 D_{S5}}{1 - D_{S5}} \quad (1)$$

Where D_{S5} represents the duty cycle of the switch S_5 . The components

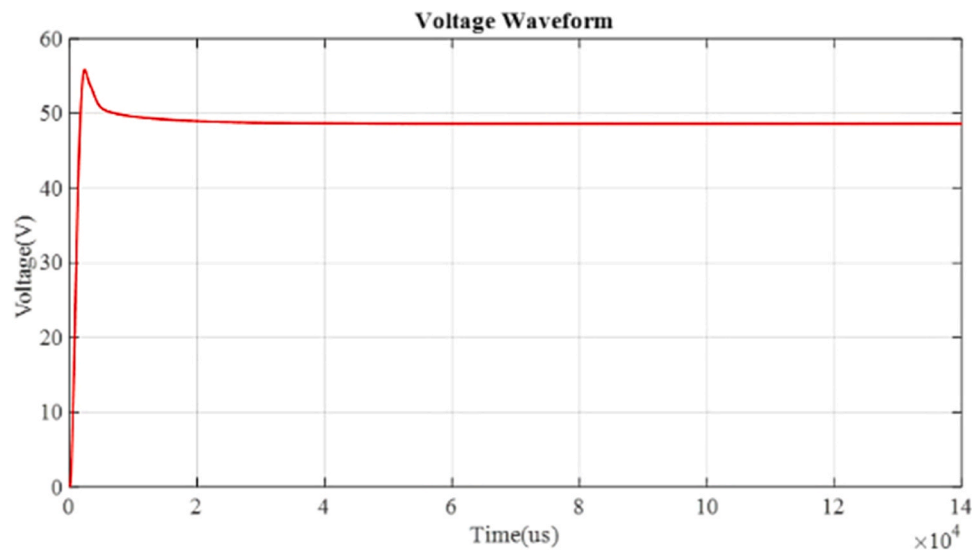


Fig. 17. Voltage waveform of charging energy storage system from energy generation system.

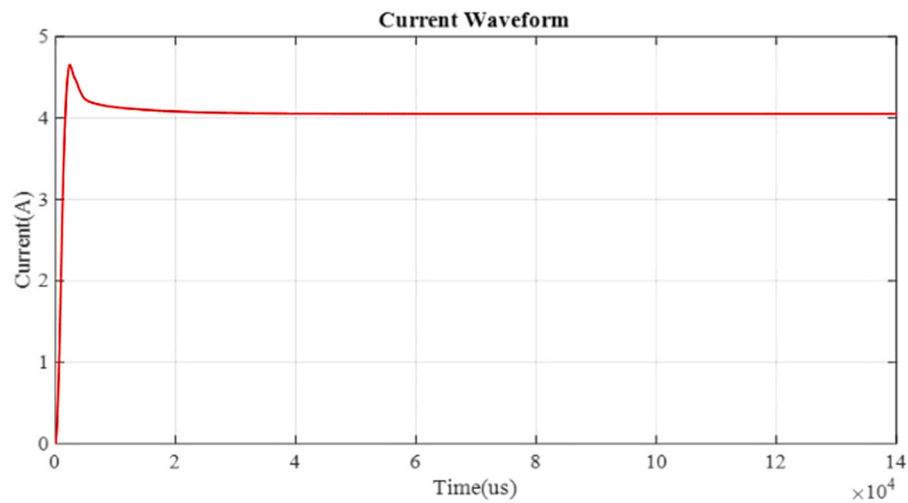


Fig. 18. Current waveform of charging energy storage system from energy generation system.

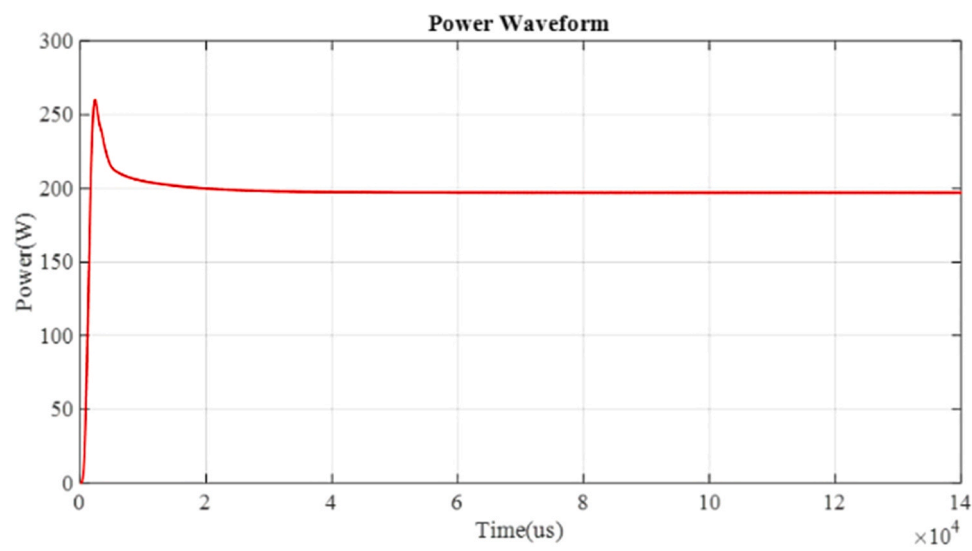
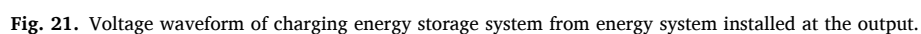
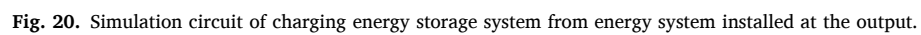


Fig. 19. Power waveform of charging energy storage system from energy generation system.



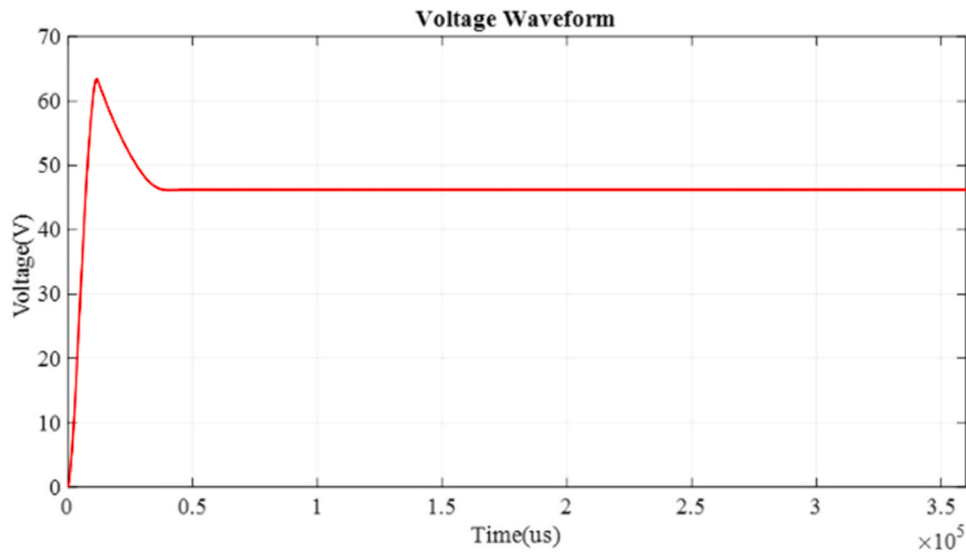


Fig. 22. Current waveform of charging energy storage system from energy system installed at the output.

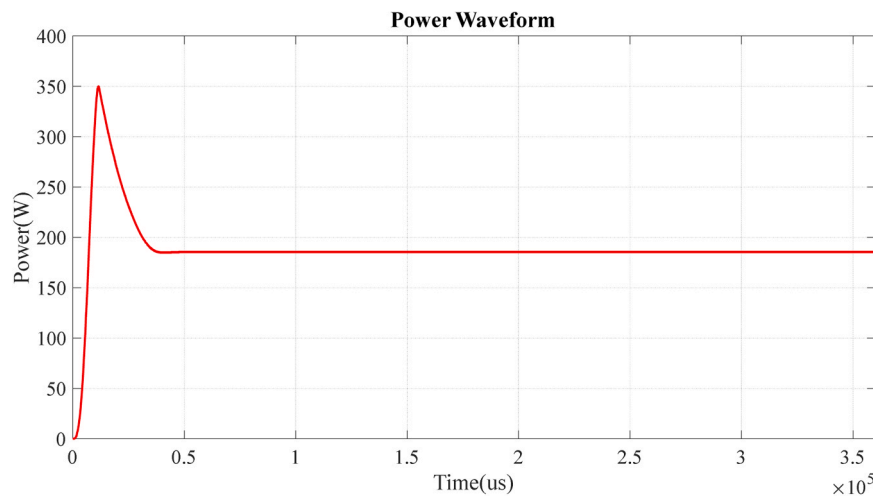


Fig. 23. Power waveform of charging energy storage system from energy system installed at the output.

C_a , D_{a1} , D_{a2} and N_{a3} compose a lossless passive snubber circuit in the upper phase of the converter. This snubber circuit employs zero voltage switching for turning off the switch S_5 . The turns ratio between N_{a1} and N_{a3} should be described such that the voltage V_{ca} across the snubber capacitor C_a reaches the output voltage V_o as:

$$n_2 \geq \frac{V_o}{V_i} \quad (2)$$

The zero-voltage switching condition for turning off the switch S_5 is obtained by satisfying the inequality mentioned in Eq. 2. Further, the location of S_5 is in between the two inductors, hence, the zero current switching condition of tuning on the switch S_5 will be attained.

In the operating mode of the converter of the energy storage system to load energy transfer, S_2 and S_3 are the switches employed in this condition. The voltage gain under this operating condition is expressed as:

$$\frac{V_{out}}{V_{in}} = \frac{1 + n_4 D_{S2}}{1 - D_{S2}} \quad (3)$$

Where D_{S2} and D_{S3} represent the duty cycles of the switches S_2 and S_3 , respectively. S_1 and S_2 will be switched in this operating condition during zero voltage switching. Turning off condition under zero voltage

switching is established by the snubber capacitors C_1 and C_2 . The energy stored in the snubber capacitors must be released before the turning on of the switch so that the soft switching mechanism would be available at the turning on condition. An active clamp circuit along with the leakage inductance energy is utilized for establishing this soft switching mechanism. The turning-on condition for zero voltage switching can be represented by the following inequality.

Energy dissipated across leakage

inductance $L_{LK} \geq$ *Energy dissipated across*

snubber capacitors $(C_1 + C_2)$ (4)

$$\frac{1}{2} L_{LK} i_{LK}^2 \geq \frac{1}{2} (C_1 + C_2) V_{ds}^2 \quad (5)$$

In the operating condition of an energy storage system charging by the energy generation system, the voltage gain and semiconductor stresses are equivalent to the synchronous buck converter. It is mandatory for the soft switching operation of the switches that the value of inductance should be designed to operate the converter in the boundary conduction mode. As a result, the inductor (L_{M2}) current will become negative to develop the condition of zero voltage switching and is

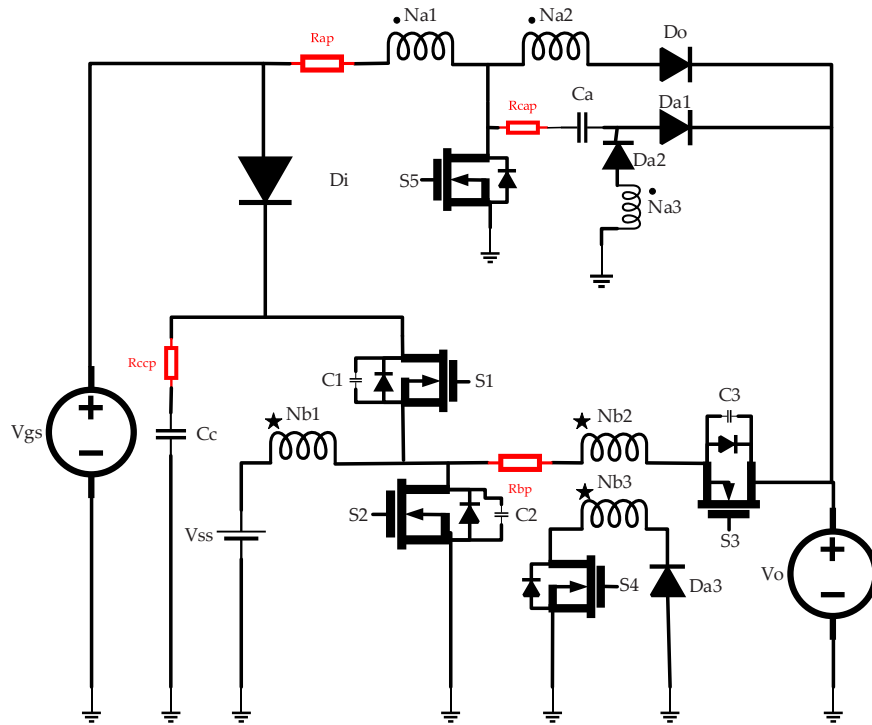


Fig. 24. Proposed circuit of multi-port DC-DC converter based on a soft-switching mechanism under parasitic resistance of coupled inductors and snubber capacitors.

Table 4

Parasitic power loss comparison under different modes of the proposed multi-port converter.

Sr. No.	Modes of Operation	$I^2 R_{cap}$ (W)	$I^2 R_{cap}$ (W)	$I^2 R_{ap}$ (W)	$I^2 R_{bp}$ (W)
1	Energy Generation System to Load Energy Transfer	-	4.6439	4.3051	-
2	Energy Storage System to Load Energy Transfer	0.1694	-	-	4.4745
3	Energy Storage System Charging from Energy Generation System	0.1694	-	-	4.4745
4	Energy Storage System Charging from Output Energy	-	-	-	4.4745

represented by the following inequality.

Energy dissipated across leakage

inductance $L_{LK} > \text{Energy dissipated across}$

snubber capacitors $(C_1 + C_2)$ (6)

$$\frac{1}{2} L_{M2} i_{LMin}^2 > \frac{1}{2} (C_1 + C_2) V_1^2 \quad (7)$$

In the operating condition of the energy storage system charging from output energy, D_{S2} , D_{S3} , and D_{S4} represent the duty cycles of the S_2 , S_3 and S_4 . The voltage gain in this operating condition is given by the following expression:

$$\frac{V_2}{V_{out}} = \frac{D_{S3}}{D_{S3} + (1 - D_{S3})(1 + n_4)} \quad (8)$$

These four operating modes of the converter help to accommodate the extensiveness of the microgrids.

4.1. Selection of Switching Frequency

The selection of switching frequency for the proposed multiport converter involves the tradeoff between component size, complexity, efficiency, and thermal performance. The basic objective behind the design of the proposed topology is to enable independent power flows under the mechanism of soft switching. This independence offers the isolation of multiple sources connected to a converter when one of the sources is experiencing an unacceptable situation. Therefore, during the modes of transferring energy from the energy generation source or the energy storage system to the load, the value of magnetizing inductance is selected in a way that a continuous conduction mode is attained. The selection of magnetizing inductance is dependent on the switching frequency which results in minimization of the current ripple and maximization of the voltage gain.

During the mode of charging of the energy storage system from the energy generation system or the output port, the value of magnetizing inductance is selected based on switching frequency in a way that the boundary conduction mode is attained.

Under the ideal conditions, the converter's mode of continuous conduction and boundary conduction are ensured by keeping the switching frequency of 50 KHz. This helps to keep the converter's inductor constant in a switching period which in turn satisfies the dead-time value of the soft switching condition (Faraji and Farzanehfard, 2021).

5. Simulation results and discussion

OrCAD is one of the utmost commonly used schematic design software for the creation of electric circuits. It is coupled with the elective OrCAD CIS for components data organization, along with the extremely integrated flows of the auxiliary engineering process. The OrCAD Capture is the utmost power for environment design which is used nowadays for the creation of the concept to production. The main use of OrCAD for electrical and electronic design is to create schematics, simulation, and PCB manufacturing. Based on its applications, OrCAD is used for the simulation of the microgrid's test converter. The parameters used in the

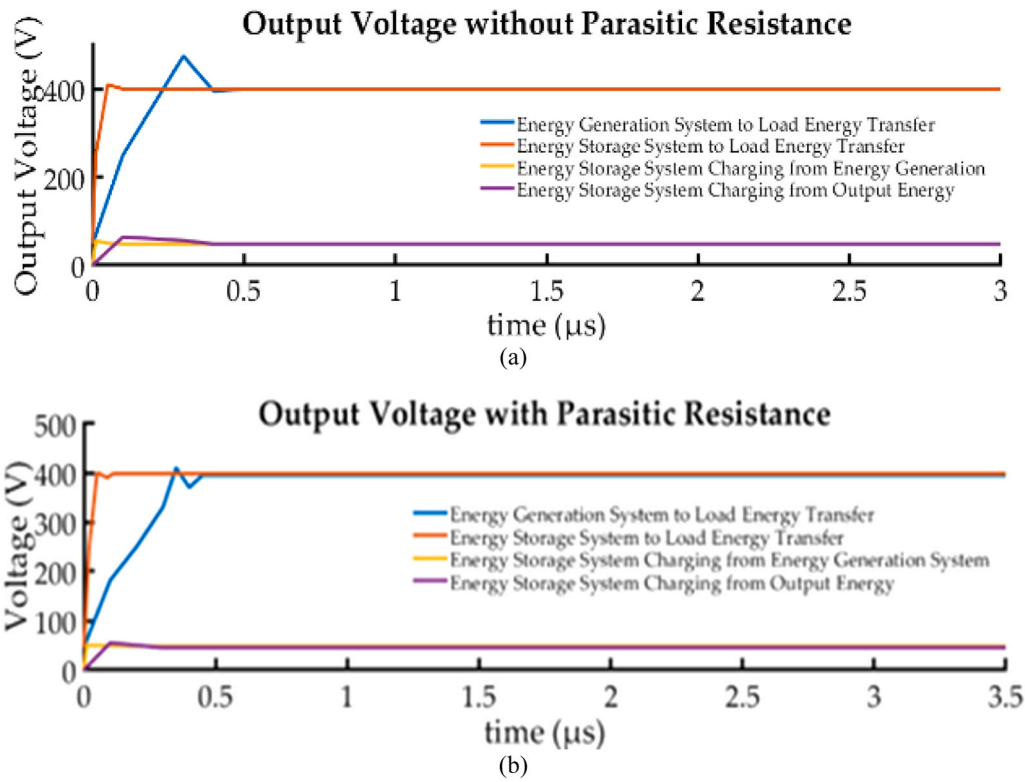


Fig. 25. The figure shows the comparison of the output voltage of the proposed multiport converter with and without parasitic resistance under different modes of operation.

Table 5

Comparison of THD under different load conditions for the proposed multiport converter for Microgrid.

Sr. No.	Mode of Operation of Multiport Converter	Load Value (Ω)	Voltage Value (V)	THD (%)
1	Energy Generation System to Load Energy Transfer	25	399.4	0.13
		50	399.7	0.28
		100	399.9	0.69
2	Energy Storage System to Load Energy Transfer	25	397.8	0.16
		50	398.6	0.67
		100	400.1	1.66
3	Energy Storage System Charging from Energy Generation System	25	47.96	0.25
		50	47.98	0.27
		100	47.99	0.28
4	Energy Storage System Charging from Output Energy	25	47.15	0.49
		50	47.97	0.49
		100	48.41	0.47

simulation results are given in Table 2.

The test converter consists of two parts. In the upper part, V_1 is connected to the energy generation system of the microgrid. The energy generation system can be photovoltaic cells and fuel cells. The energy generation system transfers energy to the DC link as the output. In the lower part, V_2 is connected to the energy storage system which charges directly from V_1 . It has a bidirectional power flow with the output. The working of the test converter is disintegrated into four different operating conditions based on the state of charge of the V_2 , the generation of energy from V_1 and the power demand of the load. The circuit diagram of the test converter is given in Fig. 2. Four different cases are analyzed depending on the four different operating conditions of the test system. The cases are described in Table 3.

5.1. Case I: energy generation system to load energy transfer

The components used in this case are given in Fig. 8. In this case, the load is given power directly from an energy generation system. It has an independent power flow path from other modes of the converter. This design acts as a boost converter and boosts the voltage in this mode. To enhance the gain of voltage, coupled inductors ($L_{12} - L_{10}$) is used. M_1 is the switch with the duty cycle of 50 % and D_1 is the output diode. A snubber circuit is used to turn off the zero-voltage switching of switch M_1 . The snubber circuit is made up of C_{20} , D_4 , D_5 , and L_{11} . L_{11} is wound on the same magnetic core as coupled inductors ($L_{12} - L_{10}$). For zero switching turn on, the switch is connected between the winding of the coupled inductor. The characteristics of voltage, current, and power in this operating case are presented in Fig. 9, Fig. 10, and Fig. 11, respectively.

5.2. Case II: energy storage system to load energy transfer

Fig. 12 represents the number of components employed in this case. In this case of the test converter, a boost converter is simulated. Coupled energy ($L_3 - L_4$) are used. The energy of the energy storage system goes to load through ($L_3 - L_4$). Switch S_3 acts as the main switch. An active clamp circuit is used for soft switching. It is made up of M_{11} and C_1 .

As for zero voltage switching turn-off, snubber capacitors C_5 and C_6 are used. Before the switch is turned on, the charge stored in capacitors is discharged through leakage inductance, so that zero voltage switching conditions of turn on can be achieved. The diode used in Case I, D_3 , and D_4 are reversed-biased and hence are separated from the upper side of the converter of the microgrid. The main waveforms of voltage, current, and power in this operating case are shown in Fig. 13, Fig. 14, and Fig. 15, respectively.

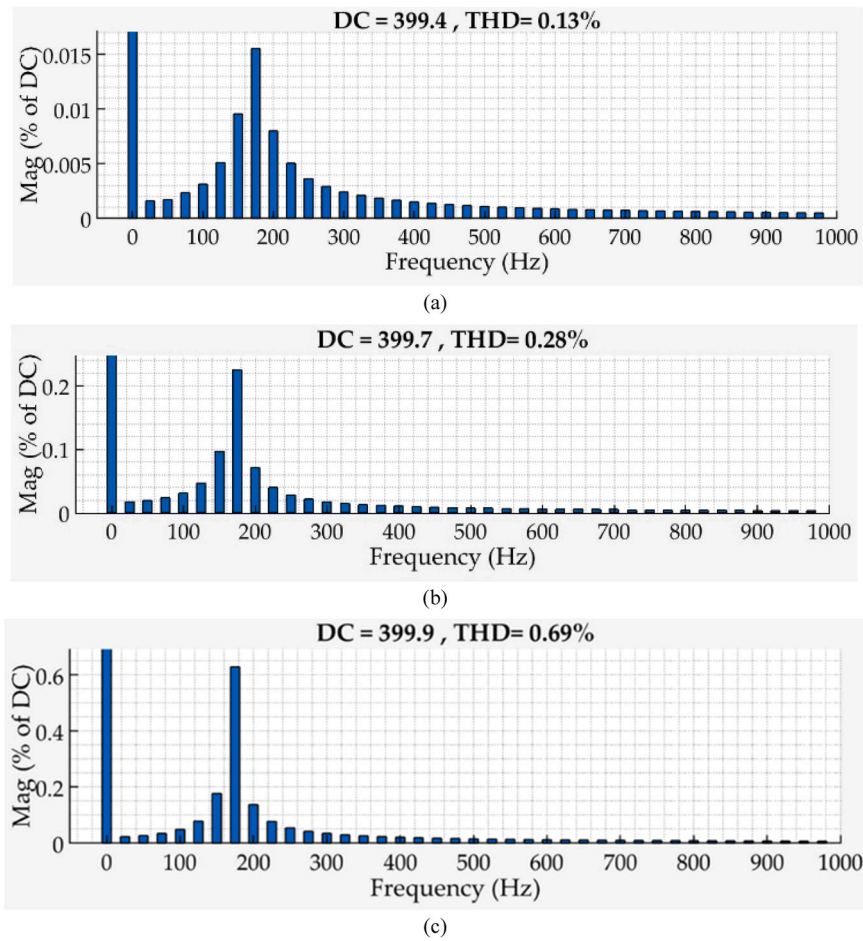


Fig. 26. The THD analysis of the proposed multiport converter at (a) 25 Ω , (b) 50 Ω , and 100 Ω under the mode of energy generation system to load energy transfer.

5.3. Case III: Energy Storage System Charging by Energy Generation System

The components employed in this operating condition are shown in Fig. 16. In this condition, the energy storage system is charged directly from the energy generation system. This operating condition is known as a buck converter. Two switches M_{10} , M_{11} , and one inductor L_4 are deployed to make a buck converter. M_{10} acts as the main switch and M_{11} as a rectifier. For zero voltage switching turn-off conditions, the snubber capacitors C_5 and C_6 are used. The value of L_4 is designed in a way that the converter works in boundary conduction mode so that zero voltage switching turn-on conditions can be achieved. The main waveforms are shown in Fig. 17, Fig. 18, and Fig. 19.

5.4. Case IV: energy storage system charging from output energy

In this operating condition, the components employed are shown in Fig. 20. The energy storage system is charged from the output by a bidirectional power transfer. The buck converter is used in this test system. The lower side of the converter acts as the bidirectional converter. The switch M_8 is the main switch and D_9 is the rectifier. To buck the output voltage, the coupled inductors L_4 – L_5 are utilized. For soft switching, D_9 and L_3 are employed. Capacitor C_4 is deployed as a filter. The main waveforms are shown in Fig. 21, Fig. 22 and Fig. 23.

5.5. Effects of parasitic resistance of capacitors and inductors

The parasitic resistance affects the proposed converter's efficiency because of expected power losses across them. To evaluate the

performance of proposed multiport converters, four different parasitic resistances are added in the design of the multiport converter as shown in Fig. 24. The parasitic resistances are chosen across the coupled inductors and snubber capacitors.

The parasitic resistances are connected in series to the coupled inductors, power switches, and snubber capacitors.

The power losses across the components of the proposed converter are given by the following expression:

$$P_{loss} = I^2 R_{parasite} \quad (9)$$

The total power loss in the proposed multiport converter that will affect the efficiency of the proposed multiport converter is given as:

$$P_{loss} = I^2 R_{cap} + I^2 R_{cap} + I^2 R_{ap} + I^2 R_{bp} \quad (10)$$

The power loss under different modes of operation of the proposed multiport converter is presented in Table 4.

Moreover, the parasitic resistance also reduces the desired output voltage increasing by acquiring more current to maintain power requirements at the output. The output voltage comparison is given in the Fig. 25.

5.6. Total harmonic distortion analysis of proposed converter under different load conditions

Total harmonic distortion (THD) analysis is necessary to validate the performance of the proposed multiport converter under multiple power flow paths and various renewable energy sources.

Therefore, to meet the standards of power quality, the value of total harmonic distortion should be as minimum as possible. In this research,

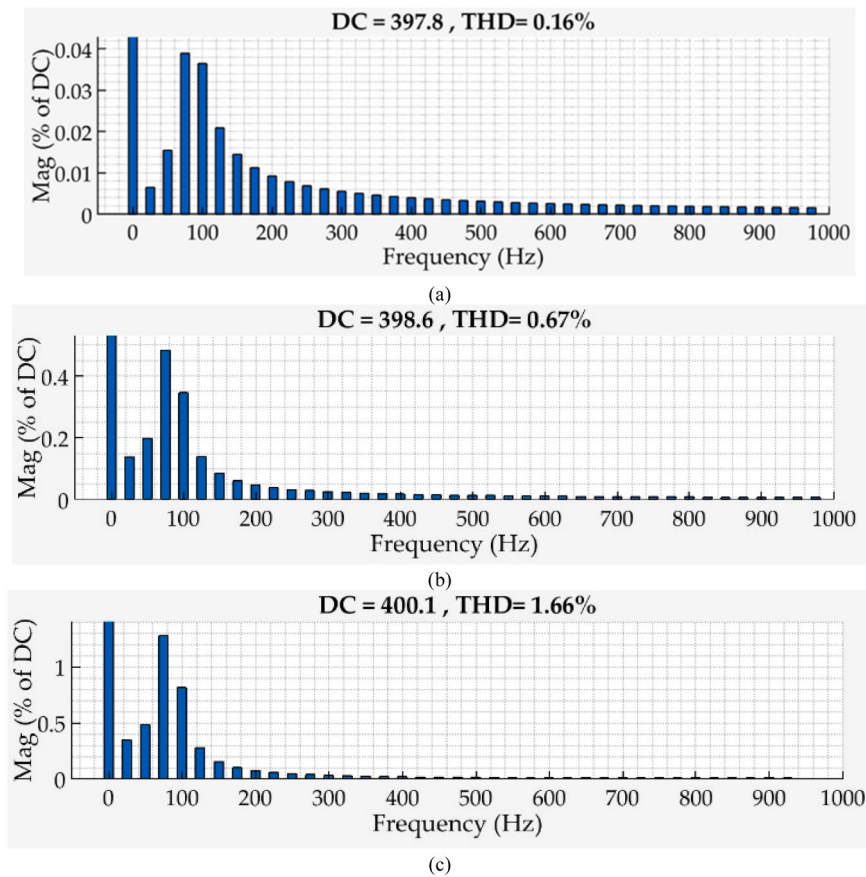


Fig. 27. The THD analysis of the proposed multiport converter at (a) 25 Ω , (b) 50 Ω , and 100 Ω under the mode of energy storage system to load energy transfer.

the operating modes of the proposed multiport converter are tested under various load conditions and the tabular results are presented in Table 5.

Moreover, the THD analysis under different modes of operation of the multiport converter is given in Fig. 26, Fig. 27, Fig. 28, and Fig. 29.

5.6.1. Conclusive remarks about THD analysis

Following are the conclusions derived from the simulation results of THD analysis of the proposed multi-port converter:

1. In the mode of the energy generation system to the load energy transfer, the value of the voltage at which the load is fed is retained. Variation in the value of the load will not affect the generation system with this proposed multiport converter. Peak and off-peak hours can easily be addressed in the microgrid without peak shaving.
2. In the mode of energy storage system to the load energy transfer, low THD not only reduces the power losses but also prolongs the life of the equipment. Hence, a stable and reliable supply is available to the sensitive loads from the energy storage system.
3. In the mode of charging the energy storage system from the energy generation system, low THD is an emblem of efficient and effective energy transfer. The life of the energy storage system is prolonged because of maintained health under proper charging.
4. The value of the voltage is maintained in the mode of charging the energy storage system from the output energy. It means that generated energy at the output from the solar or wind, is effectively stored in the batteries rather than being lost due to high THD. This feature ensures the smooth charging of the energy storage system under the source connected to output port of the proposed multiport converter.

Hence, advanced control techniques, optimal converter topologies,

and effective filtering are not required. Computational complexity and component count are reduced under this proposed multiport converter for lowering THD. IEEE standards such as IEEE 519 and IEC 61000–3–2 ensure limits on harmonic distortion to ensure power quality. A very low THD aids in complying with the power quality standards, which is necessary for both utility and customer satisfaction.

5.7. Electromagnetic interference of proposed converter

The effects of electromagnetic interference generated by the proposed multiport converter can be crucial for the integrated components and microgrids. Therefore, snubber circuits have already been added to the proposed topology to reduce the effect of electromagnetic interference. Further, the independent path flows under different modes of operation of the proposed multiport converter offer reduced generation of harmonics during switching. However, for the validation of the proposed topology, power spectrums are added in Fig. 30, Fig. 31, Fig. 32, and Fig. 33 under different modes of operation of the multiport converter. The intensity of the yellow color increases from left to right as the power of frequency components increases from 0 dB to 40 dB. The intensity of the blue color increases from right to left as the power of frequency components decreases from 0 dB to –140 dB.

The parasitic resistive plus inductive load is considered for the simulation so that better information can be gathered. In Fig. 30, the power spectrum plot is supported with the spectrogram to understand the crest and trough of power of frequency components in the case of an energy generation system to load energy transfer. Only a few peaks are identified above the 30 dB but not greater than 45 dB. The majority of the frequency components' peaks exist within the range of 0–30 dB. This highlights the minimal power losses under parasitic resistance and switching operations making the proposed model viable for this case.

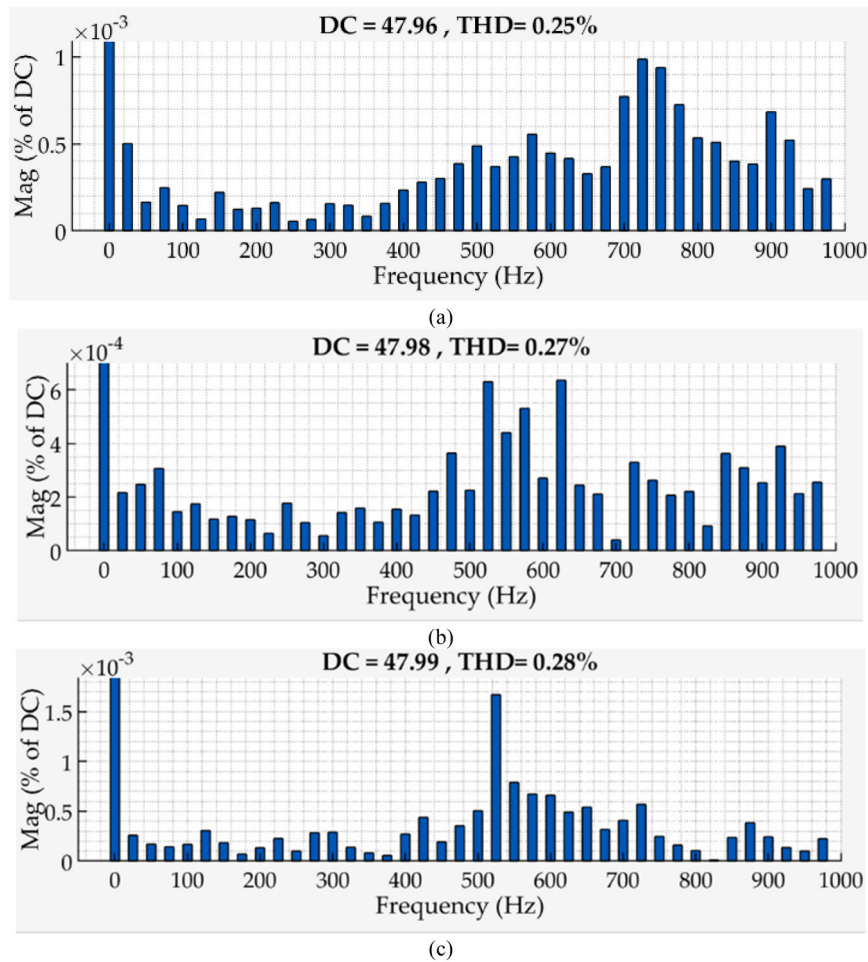


Fig. 28. The THD analysis of the proposed multiport converter at (a) 25 Ω , (b) 50 Ω , and 100 Ω under the mode of energy storage system charging from the energy generation system.

In Fig. 31, the spectrogram is plotted along with the power spectrum plot to develop an insight into the understanding of the maxima and minima of power of frequency components in the case of an energy storage system to load energy transfer. Only a few peaks are identified above the 30 dB but not greater than 40 dB. The majority of the frequency components' peaks exist within the range of 0–30 dB. This depicts the minimal power losses under parasitic resistance and switching operations making the proposed model viable for the transfer of energy from energy storage system to load.

In Fig. 32, the power spectrum is plotted along with the spectrogram to understand the crest and trough of power of frequency components in the case of charging the energy storage system from the energy generation system. Only a single peak is identified above the 30 dB. All the peaks of frequency components exist between 0 and 30 dB. This highlights the minimal power and conduction losses under parasitic resistance and switching operations. Hence, the charging of energy storage systems can be conducted without affecting the health of the storage systems.

In Fig. 33, the plot of the power spectrum is supported with the spectrogram to understand the magnitude of the power of frequency components in the case of charging the energy storage system from the source connected to the output port of the proposed multiport converter. Only a few peaks are identified above the 0 dB. All the peaks of frequency components exist between 0 and 30 dB. This highlights the minimal power and conduction losses under parasitic resistance and switching operations. Hence, the charging of energy storage systems from the source connected to the output port can be conducted without

affecting the health of the storage systems.

5.7.1. Conclusive remarks about electromagnetic interference

Following are the conclusions derived from the simulation results of the power spectrum regarding electromagnetic interference:

1. The snubber circuits play a significant role in reducing electromagnetic interference.
2. In the modes of transferring energy from energy generation system to load and energy storage system to load, a very small portion of values higher than 30 dBs are seen in the power spectrum of the output voltage. These elevated values are relatively higher than average values. This indicates negligible interference.
3. In the mode of charging the energy storage system from the energy generation system and output energy, very few values in the samples of the output voltage are greater than 0 dB which indicates the smoothness during the switching of these modes.
4. The simulation results of the proposed converter offer a chance to reduce the EMI noise somewhat without the requirement for additional filtering components. This hypothesis is quite helpful for the converter design, particularly in electric cars with constrained volume and space.

5.8. Effectiveness of hybrid energy integrated multiport converter for distributed generation enabled microgrids

The proposed microgrid's converter offers to switch to another

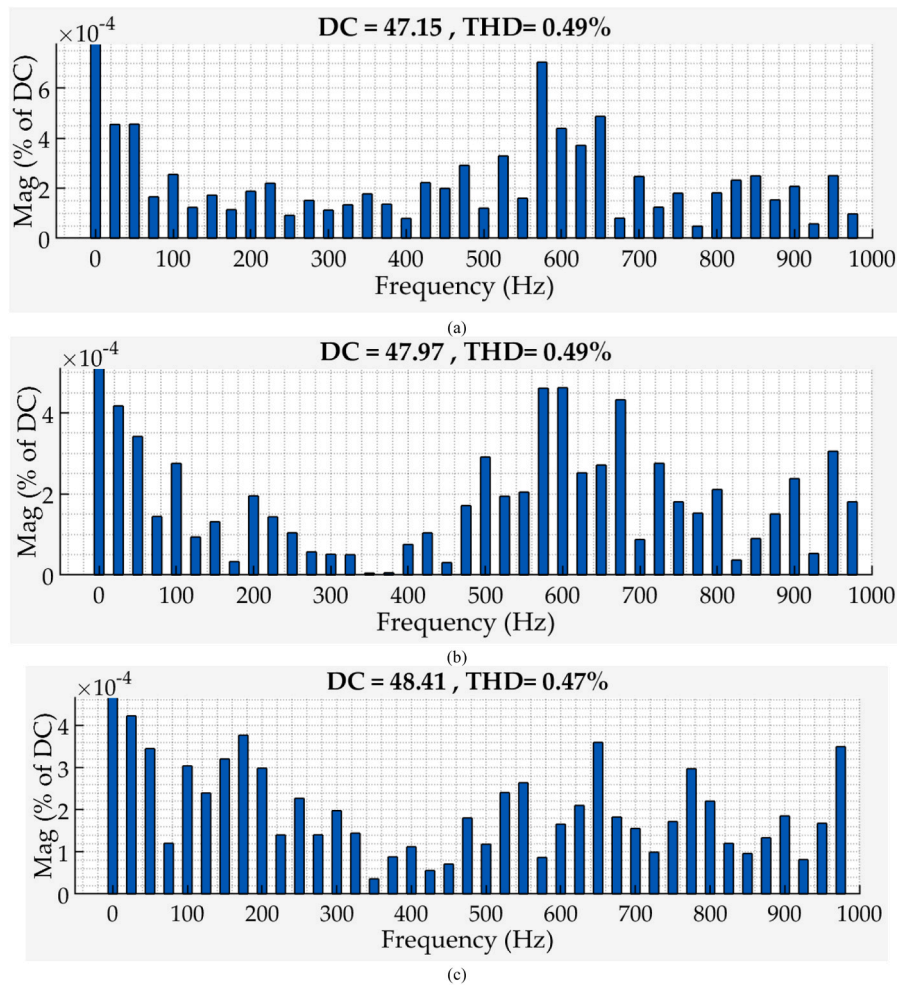


Fig. 29. The THD analysis of the proposed multiport converter at (a) 25 Ω , (b) 50 Ω , and 100 Ω under the mode of energy storage system charging from output energy.

energy source within the minimum possible time, providing a reliable supply. The performance of the proposed multiport converter is tested under two simulation scenarios. These simulation scenarios are:

- Transfer of energy from the energy generation system and the energy storage system to load
- Charging of the energy storage system from the energy generation system and output port of the suggested converter

The simulation scenario is presented in Fig. 34. The Fig. 34 provides the following information.

- The blue line indicates the switching between the energy generation system and the energy storage system to ensure a reliable supply at the load side. At point 0.3 μ s, the energy generation system is tripped, and the energy storage system is made available, at point 0.35 μ s, the energy storage system is made tripped because of the availability of the energy generation system. The energy generation system attains the steady state at 0.75 μ s. It means that 0.4 μ s is enough to feed the load at a steady voltage of 400 V.
- The red line indicates the charging of the energy storage system from the energy generation system and energy system available at the output port of the multiport converter. Charging is initiated from the energy generation system in less than 0.01 μ s. If the energy generation system is tripped at 0.2 μ s, then charging is done from the output energy. It has attained a steady state of 48 V in less than 0.15 μ s. This results in the longevity of the life of storage systems by making their

charging cycle smooth under the renewable energy generation system.

6. Validation of simulation results by mathematical calculations

The different cases of operation of the microgrid's test converter are validated through mathematical calculations.

6.1. Case I: Energy Generation System to Load Energy Transfer

V_{out} is the output voltage and V_{in} is the input voltage of the energy generation system. S_5 is the main switch with the duty cycle D_{S5} . By using the volt-second balance on L_{M1} , and neglecting the leakage inductance effect, the voltage gain is:

$$\frac{V_{out}}{V_{in}} = \frac{1 + n_1 D_{S5}}{1 - D_{S5}}$$

$$\frac{400}{96} = \frac{1 + (2.1)(D_{S5})}{1 - D_{S5}} \quad (11)$$

$$4.16 - 4.16D_{S5} = 1 + 2.1D_{S5}$$

$$3.167 = 6.26D_{S5}$$

$$D_{S5} = 0.505 = 50.5 \%$$

The continuous conduction operation is achieved by selecting the value of L_{M1} . So, for the nominal output power of the microgrid's

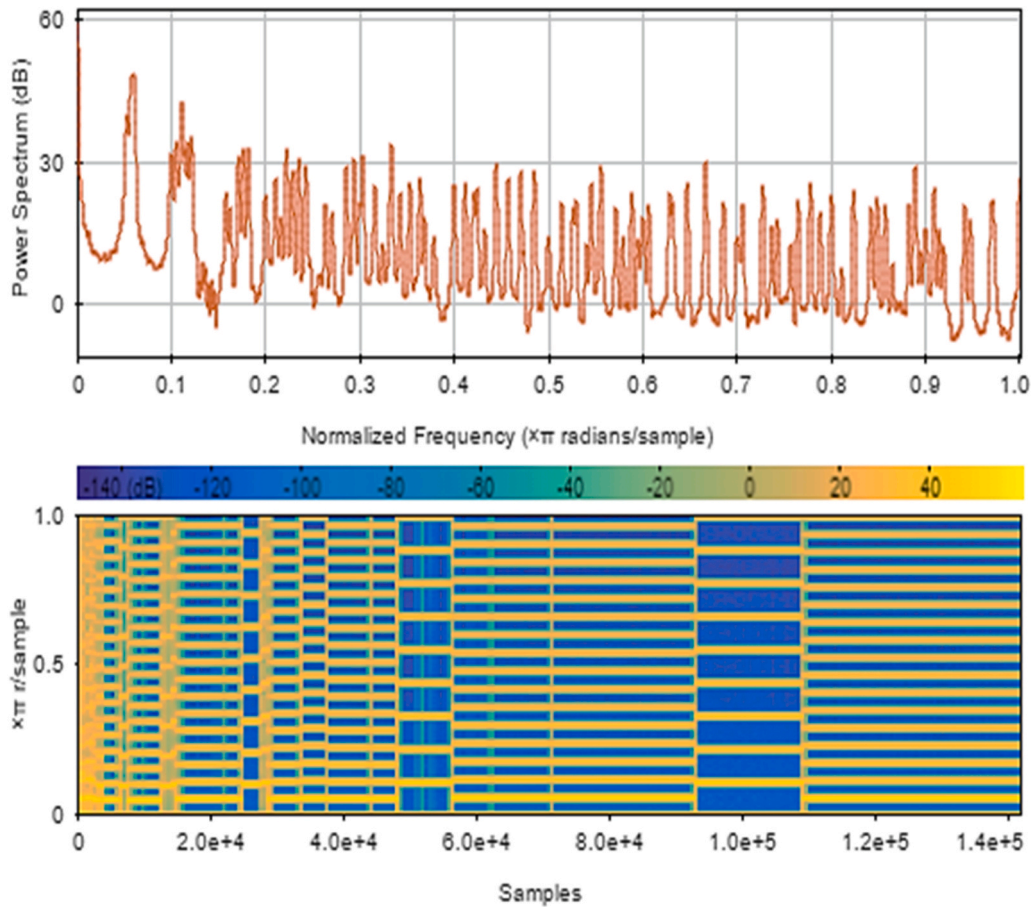


Fig. 30. The power spectrum analysis of the proposed multiport converter under the mode of energy generation system to load energy transfer.

converter, the average of magnetizing inductance current $I_{LM1(Avg)}$ and the average of ripple $\Delta I_{LM1(Avg)}$ are attained by following expressions, respectively.

$$I_{LM1(Avg)} = \frac{2 \cdot (1 + n_1 D_{SS}) \cdot I_o}{(1 - D_{SS})} = \frac{2 \cdot (1 + n_1 D_{SS}) V_o}{(1 - D_{SS}) \cdot R_L} \quad (12)$$

$$I_{LM1(Avg)} = \frac{2 \cdot (1 + (2.1 \cdot 0.505)) \cdot (400)}{(1 - 0.505) \cdot (800)} = 4.016 A$$

$$\Delta I_{LM1(Avg)} = \frac{V_1 D_{SS}}{L_{M1} \cdot f_s} \quad (13)$$

$$\Delta I_{LM1(Avg)} = \frac{(96)(0.505)}{(577e - 6)(50e3)} = 1.6804 A$$

The minimum value of magnetizing inductance is found by substituting Eq. 12 into Eq. 13 and using Eq. 11, in continuous conduction operation as:

$$L_{M1} \geq \frac{D_{SS} \cdot (1 - D_{SS})^2 \cdot R_{L(max)}}{4 \cdot (1 + n_1 D_{SS})^2 \cdot f_s} \quad (14)$$

$$577 \mu \geq \frac{(0.505)(1 - 0.505)^2 R_{L(max)}}{4 \cdot (1 + (2.1 \cdot 0.505))^2 \cdot (50e3)}$$

$$577 \mu = \frac{0.1237 R_{L(max)}}{849132.05}$$

$$R_{L(max)} = 3960.785 \Omega$$

The minimum amount of load power will be calculated at maximum load resistance $R_{L(max)}$. For zero voltage switching, the turning-off condition for switch S_5 is established using the lossless passive

snubber circuit. This snubber circuit consists of components such as C_a , D_{a1} , D_{a2} and N_{a3} . In this circuit, C_a is the snubber capacitor.

$$n_2 \geq \frac{V_o}{V_1} \quad (15)$$

$$n_2 \geq \frac{400}{96} \geq 4.16$$

Eq. 13 satisfies the zero-voltage switching turning-off condition for the switch S_5 . In between the two inductors, the main switch S_5 is connected to achieve the zero current switching turning-on condition of the switch S_5 .

6.2. Case II: energy storage system to load energy transfer

In this operation case, the switch S_2 and the switch S_3 are switched with duty cycles D_{S2} and D_{S3} respectively. Voltage gain is evaluated by removing the effects of the leakage inductance and by using the volt-second balance in magnetizing inductance (L_{M2}) as shown in Eq. 16. Also, Eq. 17 as;

$$\frac{V_{out}}{V_2} = \frac{1 + n_2 D_{S2}}{1 - D_{S2}} \quad (16)$$

$$\frac{400}{48} = \frac{1 + 3D_{S2}}{1 - D_{S2}}$$

$$8.333 - 8.33D_{S2} = 1 + 3D_{S2}$$

$$7.33 = 11.33D_{S2}$$

$$D_{S2} = 0.6469 = 64.69 \%$$

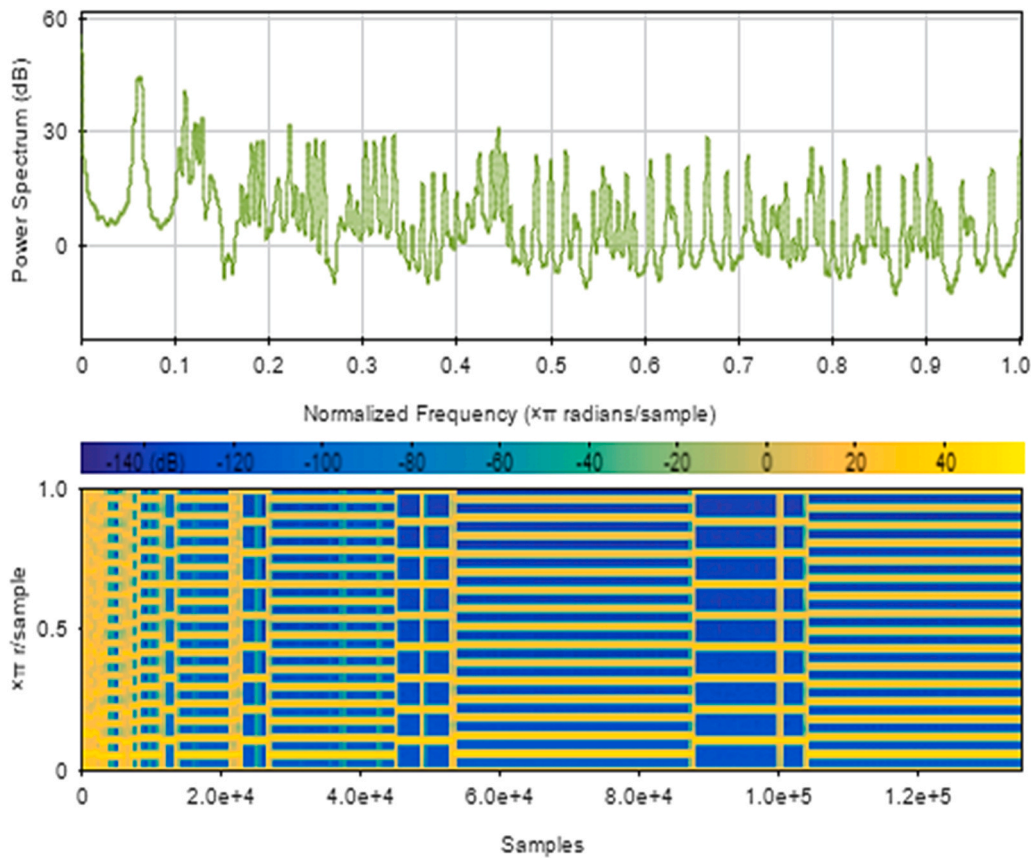


Fig. 31. The power spectrum analysis of the proposed multiport converter under the mode of the energy storage system to load energy transfer.

$$L_{M2} \geq \frac{D_{S2} \cdot (1 - D_{S2})^2 \cdot R_{L(\max)}}{4 \cdot (1 + n_1 D_{S2})^2 \cdot f_s} \quad (17)$$

$$57\mu \geq \frac{0.6469 \cdot (1 - 0.6469)^2 (3960.785)}{4(1 + (3 \cdot 0.6469))^2 (50e3)}$$

S_1 and S_2 are the switches which are under zero voltage switching conditions. To achieve this, snubber capacitors (C_1 and C_2) are used to offer zero voltage switching turn-off conditions. Before the switches are turned on, the energy store should be discharged in the snubber capacitors to provide soft switching at turn on. To attain that condition by using active energy leakage inductance and clamp circuit.

Eq. 16 is for finding conditions for which D_{bc} becomes reverse biased in this operation case:

$$\frac{V_2}{1 - D_{S2}} \geq V_1 \quad (18)$$

$$\frac{48}{1 - 0.6469} \geq 96$$

$$135.93 \geq 96$$

6.3. Case III: energy storage system charging by energy generation system

The synchronous buck converter is used to charge the energy storage system from the energy generation system in this operation case. The voltage gain of the synchronous buck converter is equivalent to the stresses on the semiconductor. L_{M2} acts as an inductor of the buck converter. The condition of boundary conduction is attained by designing the specific conductor in a way that the condition of soft switching can be obtained.

$$L_{M2} = \frac{1}{2} \cdot \frac{V_1 - V_2}{P_{V2_{\max}}} \cdot \frac{V_2^2}{V_1} \cdot \frac{1}{f_{sw}} \quad (19)$$

$$L_{M2} = \frac{1}{2} \cdot \frac{(96 - 48) \cdot 48^2}{200} \cdot \frac{1}{96 \cdot 50e3} = 57\mu H$$

Where f_{sw} is switching frequency and $P_{V2_{\max}}$ is the maximum requirement of power from V_2 . From Eq. 19, the value of the inductor fulfills the soft-switching condition which depends on the values of V_1 and V_2 . The value of the inductor gives zero voltage switching in the boundary conduction state resulting in the negative current of the inductor. Therefore, the minimum and maximum current of the inductor must be highlighted. The mathematical relations of L_{M2} value, the minimum current (I_{Min}) and the peak current (I_{Peak}) are expressed as:

$$I_{Peak} = I_{Load} + \Delta I_{LM2} \quad (20)$$

$$I_{Peak} = 0.5 + 4.8 = 5.3A$$

$$I_{Min} = I_{Load} - \Delta I_{LM2} \quad (21)$$

$$I_{Min} = 0.5 - 4.8 = -4.3A$$

$$\Delta I_{LM2} = \frac{1}{2} \cdot \frac{V_1 - V_2}{L} \cdot \frac{V_2^2}{V_1} \cdot \frac{1}{f_{sw}} \quad (22)$$

$$\Delta I_{LM2} = \frac{1}{2} \cdot \frac{(96 - 48) \cdot 48^2}{50} \cdot \frac{1}{96 \cdot 50e3} = 4.8A$$

$$I_{V2} = \frac{P_{V2}}{V_2} = \frac{200}{48} = 4.166 \text{ A} \quad (23)$$

So, to make sure that the current of the inductor for all ranges of output power should be negative, the inductor's value should be set below $57\mu H$. If the value of the inductor is reduced, the I_{Peak} value is increasing. Therefore, the rate of maximum current in semiconductor components is evaluated based on the minimum value of the inductor.

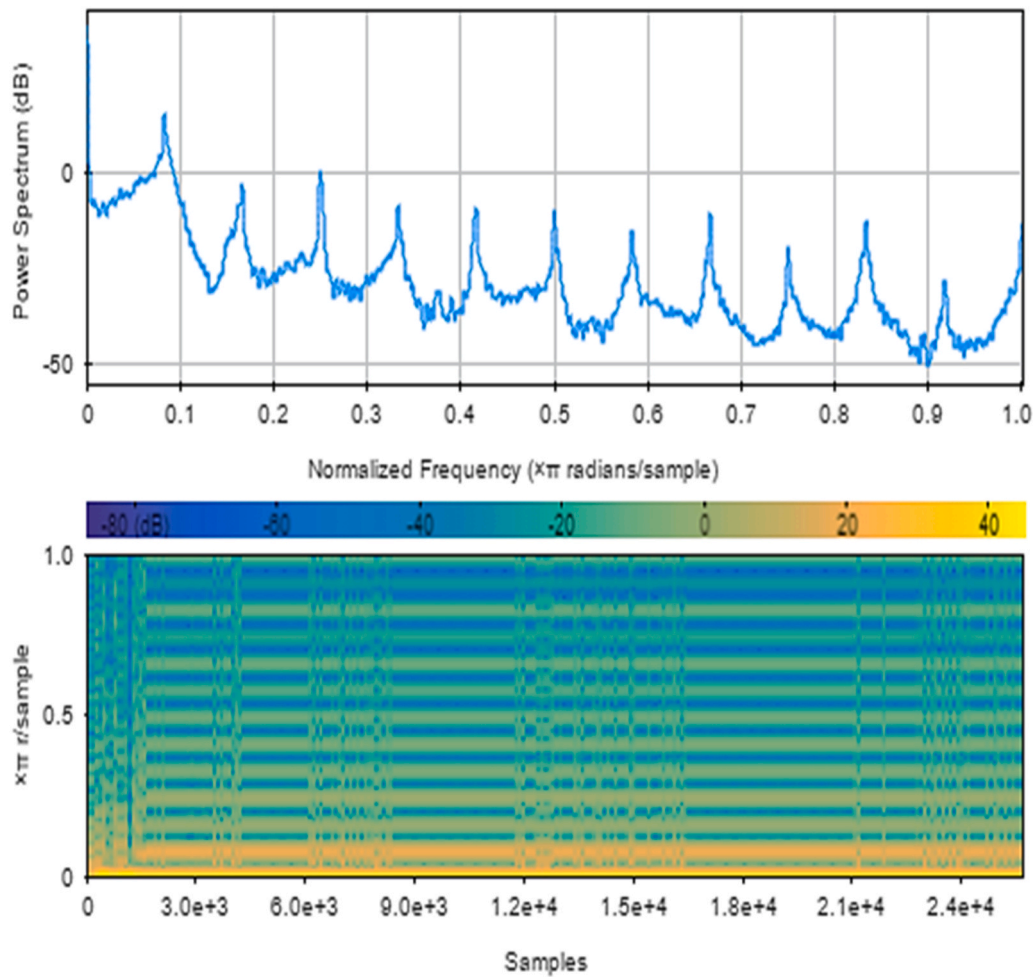


Fig. 32. The power spectrum analysis of the proposed multiport converter under the mode of the energy storage system charging from the energy generation system.

The body diode S_3 should not conduct during this operating mode. This is ensured by the Eq. 24.

$$V_0 > (V_1 - V_2)n_{b4} \quad (24)$$

$$400 > (96 - 48)(4.7)$$

$$400 > 225.6$$

6.4. Case IV: energy storage system charging from output energy

In this operation case, the switches are S_2 , S_3 , and S_4 with duty cycles D_{s2} , D_{s3} , and D_{s4} respectively. The voltage gain in this operation case is evaluated by removing the effects of leakage inductance and by implementing a volt-second balance across L_{M2} . Voltage gain is expressed mathematically in Eq. 25. The continuous conduction setup is established by the value of L_{M2} and this value is expressed in Eq. 26. The snubber capacitors are C_1 and C_2 . The evaluation of C_3 needs extra consideration and is given in (Farajdadian et al., 2024).

$$\frac{V_2}{V_{out}} = \frac{D_{s3}}{D_{s3} + (1 - D_{s3})(1 + n_4)} \quad (25)$$

$$\frac{48}{400} = \frac{D_{s3}}{D_{s3} + (1 - D_{s3})(1 + 4.7)}$$

$$D_{s3} = \frac{48(1 + 4.7)}{(400 + 48 * 4.7)} = 0.4373 = 43.73\%$$

$$L_{M2} \geq \frac{[D_{s3} + (1 - D_{s3})(1 + n_4)] \cdot (1 - D_{s3})R_{L(Critical)}}{2 \cdot (1 + n_4) \cdot f_s} \quad (26)$$

$$L_{M2} \geq \frac{[0.4376 + (1 - 0.4373)(1 + 4.7)] \cdot (1 - 0.4373)(R_{L(Critical)})}{2 \cdot (1 + 4.7) \cdot (50e3)}$$

6.5. Significance of validation by mathematical calculations

Validation is a crucial step in scientific and engineering research. The performance and generalizability of the microgrid's multiport converter are assessed by the behavior observed under the assignment of different values to input, output, and its effective components. As a result, quantitative and qualitative assessments are obtained depicting the wellness of the proposed multiport converter and its fitness for the intended purpose. Based on the mathematical calculations, it is found that the mathematical results are very close to the observed results under different simulation scenarios. Mathematical calculations of the proposed microgrid's multiport converter provide the following significant information:

- The turns ratios of coupled inductors are acquired that enable the zero-voltage switching under different operating modes of the proposed multiport converter.
- The proposed multiport converter operates in continuous conduction mode and boundary conduction mode. The value of magnetizing inductance decides the conduction mode of the proposed model. The

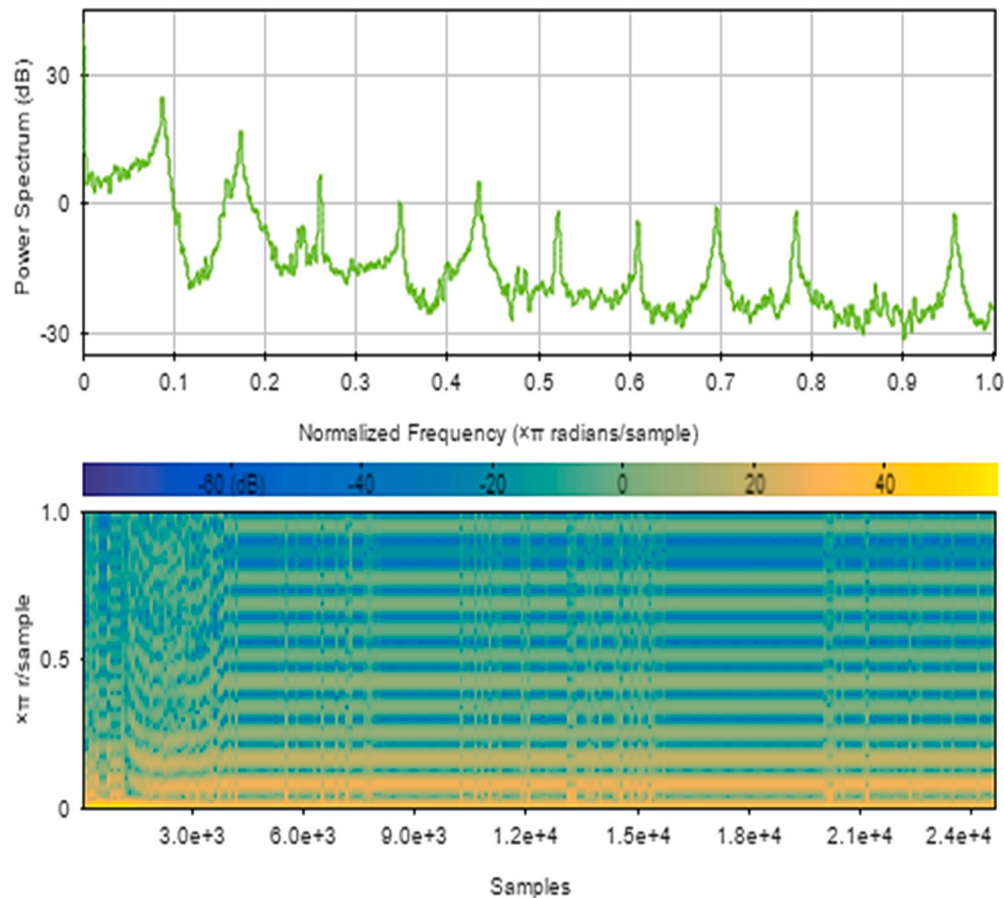


Fig. 33. The power spectrum analysis of the proposed multiport converter under the mode of energy storage system charging from output energy.

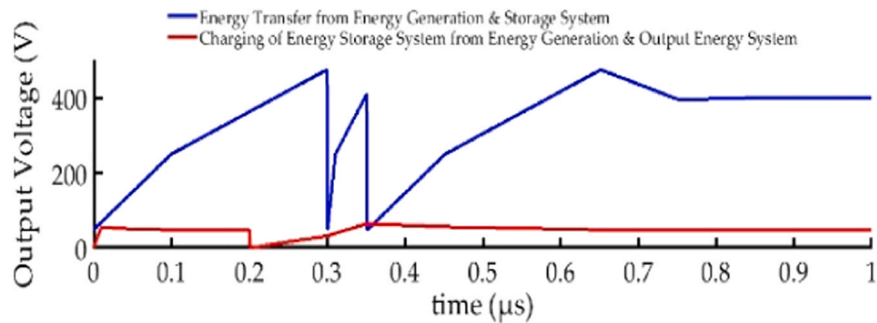


Fig. 34. The simulation scenario validating the effectiveness of a hybrid energy integrated multiport converter.

Table 6
Tabular comparison of simulated and mathematical values of the proposed multiport converter.

Modes	Parameters	Simulated Value	Mathematical Value	RMS Error	Mean Absolute Error	Error (%)
Energy Generation System to Load Energy Transfer	Voltage	398.942 V	400 V	1.058	1.058	0.2625
	Power	200.081 W	200 W	0.081	0.081	0.0405
	Current	0.5265 A	0.5 A	.0265	0.0265	5.30
Energy Storage System to Load Energy Transfer	Voltage	398.425 V	400 V	1.944	1.944	4.22
	Power	198.646 W	200 W	1.354	1.354	0.677
	Current	0.498 A	0.5 A	0.002	0.002	0.4
Energy Storage System Charging by Energy Generation System	Voltage	48.699 V	48 V	0.699	0.699	1.456
	Power	197.496 W	200 W	2.504	2.504	1.252
	Current	4.0572 A	4.16 A	0.102	0.102	2.471
Energy Storage System Charging from Output Energy	Voltage	46.248 V	48 V	1.752	1.752	3.65
	Power	185.662 W	200 W	14.33	14.33	7.169
	Current	4.0147 A	4.16 A	.1453	.1453	3.492

Table 7

Performance comparison of the proposed multiport converter with the existing multiport converters.

Sr. No.	Parameters	Existing Multiport Converters (Saritha et al., (2023), (Chen et al., 2013), (Hintz et al., 2015))	Proposed Multiport Converter
1	Soft Switching	Soft switching is obtained with the maximum number of switches, resulting in low voltage gain.	Soft switching is acquired with five switches, resulting in high voltage gain.
2	Number of Switches	Hard Switching is done with 4 and 5 switches but Soft Switching is acquired through 6 switches	Soft switching is acquired through five switches.
3	Nature of Inductors	Mostly single inductor cores are used.	Coupled inductor cores are used to facilitate high power transfer density and generation of multiple output voltages from a single input source
4	Voltage Gain	Not a significant improvement in voltage gain is observed.	Relatively high voltage gain is observed resulting in reduced current ripple and stable operation under renewable energy sources
5	Independence of Power Flows Path	Mostly independent power flow path is observed between input and output but is not tested under renewable energy generation systems and energy storage system	Independent power flow path is observed in all the modes of operation because of coupled inductors, resulting in a smooth transition among different sources of renewable energy and storage systems.

value of the magnetizing inductor is acquired through mathematical calculations.

- The turning-off condition is established by snubber capacitors at zero voltage switching. To provide soft switching at the turning-on condition, energy stored in snubber capacitors must be discharged before the turning on of the switches. The discharged energy must be tracked to leakage inductance. The value of leakage inductance is assessed by mathematical analysis.
- The bidirectional power flow can be obtained by assessing the relationship of voltage between source and load. This relationship is assessed and validated by mathematical calculations.

6.6. Comparison of simulation and mathematical analysis for the validation of proposed multiport converter of microgrids

The validation of the proposed multiport converter is proven with the help of the comparison of simulated and mathematical values. These values are measured and calculated under different modes of operation of multiport converter. Minimum error in the values of measurements conforms to its validation for microgrids and its integration with distributed generations. Table 6 provides a detailed comparison of measurements and calculations of the proposed multiport converter.

The difference between the simulated and mathematical values of power is maximum in the case of charging the energy storage system from the output port i.e., 7.169 %. It can be mainly due to the incorporation of the parasitic resistance in the simulation circuit which was not considered for the mathematical analysis. Further, in the mathematical analysis, lossless passive snubber circuits are considered. Hence this table provides a comparative analysis of the ideal and realistic environment for the operation of the proposed multiport DC-DC converter.

7. Performance comparison of proposed multiport converter with existing multiport converters of microgrids

The proposed multiport converter excels in performance because of its effective handling capacity of multiple renewable energy sources simultaneously. The adaptability of the proposed converter is significant from its attaining steady-state conditions under switching of different operating conditions. Development of steady state conditions within the minimum possible time not only maximizes the efficiency but also reduces the losses resulting in increasing the safety margins of sensitive power electronic components. Some of the performance parameters are compared with the existing converters and are presented in Table 7.

Fig. 34 presents and compares the waveforms of efficiency of the proposed multiport converter for various operating modes and output power levels with the existing multiport converters. This Fig. 35 depicts the superiority of the proposed converter over (Yu and Kwasinski, 2013) and (Faraji et al., 2019) and has high efficiency across a wide output power range.

8. Possible future applications of proposed multiport converter for microgrids

Multiport converters would be the most promising component of future power systems based on distributed generation-enabled microgrids. Therefore, the following are the advantages identified for the future:

8.1. Charging stations of electric vehicles (EVs)

With the advancement in the technology of electric vehicles, charging stations would be required throughout the world. Integration of renewable energy sources, conventional grids, and energy storage systems can be made possible with this proposed multiport converter resulting in independence in bidirectional power flow and stability of grids.

8.2. Integration of Renewable Energy Sources (RESs)

Smooth integration and reliable control and management of power from various renewable energy sources can be attained with the deployment of the proposed converter. This multiport converter provides independent paths of power flow with reduced component strategy resulting in less switching losses and reduced complexities.

8.3. Provision of Energy Storage Systems (ESSs)

The proposed multiport converter enables the charging and discharging of energy storage systems without affecting the power flow for utilization. Hence, microgrids can be beneficial not only for industrial and commercial purposes but also for telecommunication information, and technology.

8.4. Grid-support capability

With the aid of the proposed multiport converter, microgrids can ensure not only voltage reliability but also participate in demand side management and demand response in a minimum time. This helps bring resilience to the energy infrastructure not only for the rural areas but also for the congested load locations.

In developing countries like Pakistan, decentralization of energy can be introduced with the help of multiport converters. Remote areas not connected to the centralized energy infrastructure can be electrified with the help of multiport converters by the integration of locally available renewable energy sources and storage mechanisms. Distributed generation provisions with the aid of multiport converters not only reduce power outages in urban areas but also enhance the resilience of

COMPARISON OF EFFICIENCY UNDER DIFFERENT LOAD CONDITIONS

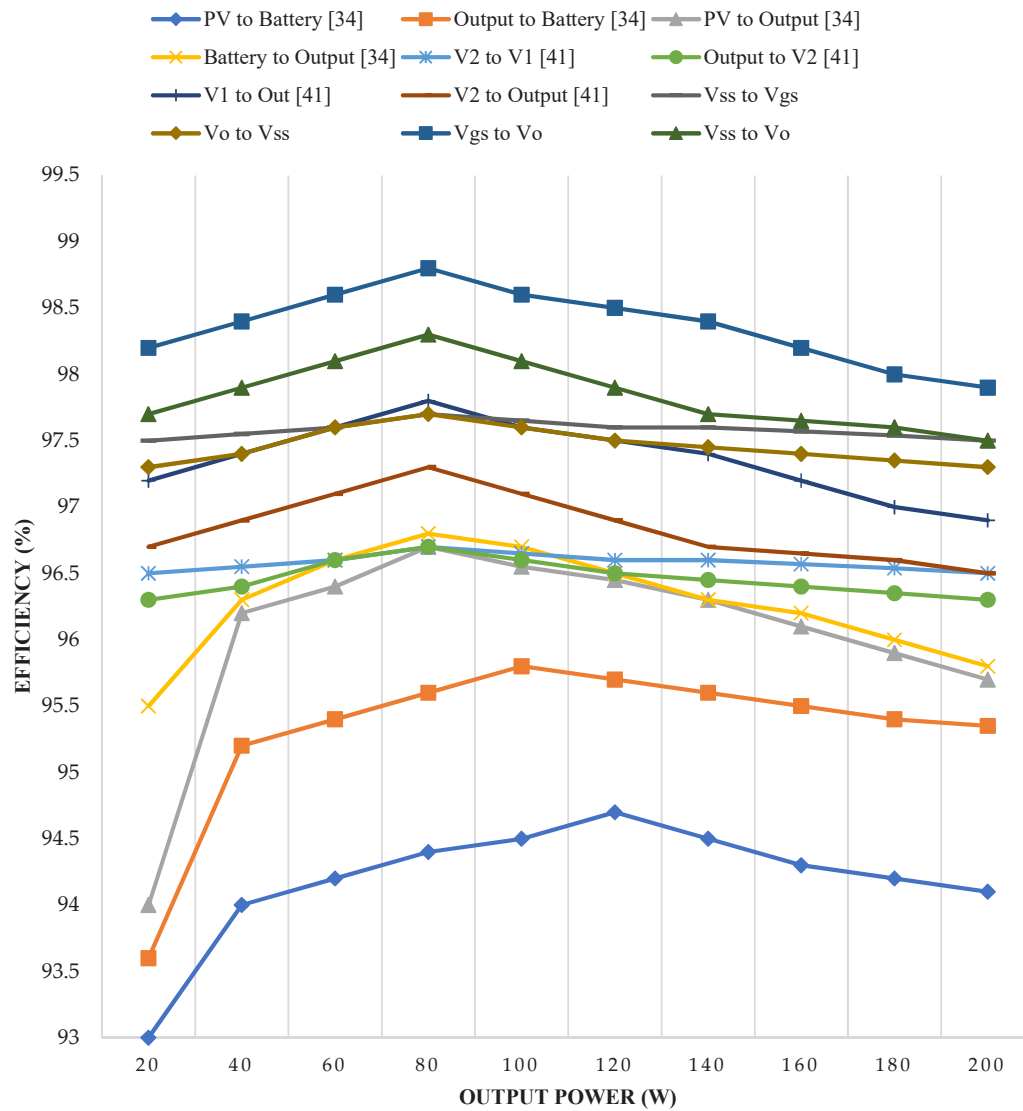


Fig. 35. The superiority of the proposed multiport converter over the existing multiport converter under various modes of operation and at different loads.

energy networks. Moreover, emergency conditions can be sorted with the help of multiport converter-based microgrids in an economical way. Technology transfer and capacity building not only promote a sense of responsibility locally but also restrict mass migration in search of standard living.

8.5. Implication and challenges regarding community involvement

Community involvement in microgrids can have several implications and challenges, both positive and potentially hindering. Microgrids give control to residents over their energy resources, resulting in a prevalent sense of ownership and education. Further, it stimulates local economic development by creating jobs for installation, maintenance, and operation. Lower energy bills can free up resources for other projects in the community. Microgrid development can be expensive, resulting in disagreement in developing countries because of too much effort required to secure funding and feasibility. Technically sound experts are required for the microgrid's initiatives. Lack of expertise might cause operational difficulties and even safety risks in communities. Moreover, it is possible that community microgrids may not be supported by the

existing regulatory frameworks. Navigation of new regulatory thinking can be time-consuming and may require extensive alteration in energy policy.

8.5.1. Proposed multiport converter – a community-friendly solution

Following are the significant benefits offered by the proposed multiport converter in the community involvement towards microgrids:

- The precise regulation of voltage at the output in the proposed converter ensures not only the stability of the microgrids but also promises longevity of the loads connected to it. This enhances confidence in the community for initiating the step towards microgrid installation.
- Social equality may be promoted by the proposed multiport converters in the microgrids. The proposed converter can optimize the distribution of energy with the intention of equitable access and benefits for all the community members connected to it.
- Communication less switching can be obtained with this multiport converter. This enables the absorption of all the power quality issues

created by one of the members by a smooth transition towards another available source.

- The proposed multiport converter offers both grid-connected and island modes of the microgrids. During periods of grid outages and disasters, the proposed multiport converter enhances the community's resilience during emergencies by shifting the load to another source.
- Community members can directly sell extra energy with one another using smart contracts or a blockchain-based system, establishing a peer-to-peer grassroots-level marketplace. By automating the invoicing and payment procedures, smart contracts may offer a reliable and frictionless peer-to-peer trading environment.

9. Conclusion

In this research, the microgrid's multiport converter is designed and analyzed under its various modes of operation so that its validity can be justified with the integration of various renewable energy sources and energy storage systems. Moreover, its work is further proven with the help of mathematical analysis. The proposed multiport converter is based on a soft switching mechanism resulting in reduced switching losses and independence of power flows for hybrid energy applications. Five switches are employed in this research model instead of six switches for soft switching. The soft switching condition is established by the management and tracking of released energy from the snubber capacitors. The released energy is utilized by the leakage inductance whose value is assessed through mathematical calculations. The tracking of released energy helps to minimize the voltage spikes and current harmonics. As a result, the power quality of the proposed multiport converter is improved during inter-switching. The high voltage gain of 8.3 times the input voltage with 96 % efficiency offers reliability under the variable nature of renewable energy sources. Further, it is proved from the simulations that attainment of the steady state within the minimum possible time reduces the current ripple, resulting in a reduced heating burden on sensitive power electronic circuitries. The study of the effects of parasitic resistance, total harmonic distortion, and electromagnetic interference makes it an ideal candidate for microgrids and distributed generation for rural and urban communities. Moreover, the proposed multiport converter offers provision of the energy storage system. The energy storage system charging is carried out from the available renewable energy source and any energy source connected to the output without any fear of disturbance to other energy applications.

9.1. Future direction

In the future, the following potential areas regarding the proposed multiport converter for microgrids can be explored:

- The maximum power point tracking (MPPT) is used to optimize the output power generated from renewable energy sources. The integration of MPPT with the proposed multiport converter can be studied to make it more scalable and adaptable with the varying renewable energy sources.
- The on-grid and off-grid modes of microgrids can be established with the proposed multiport converter. This will help to analyze the behavior with the existing power networks and remote load locations. Further, the design can be adjusted according to the requirements under these modes of microgrids.
- Microgrids have to adjust their capacity in response to shifts in the demand of energy consumed or the addition of new energy sources frequently. The versatility of the multiport converter can be studied to make the microgrid scalable, enabling the addition or removal of components without requiring a major redesign.

CRediT authorship contribution statement

Ali Altalbe: Writing – review & editing, Validation, Supervision, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Ali Raza:** Writing – review & editing, Writing – original draft, Validation, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis. **Rabia Arshad:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Raheel Muzzammel:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

"I, as the corresponding author, hereby declare that there are no competing interests to disclose pertaining to this manuscript."

Data Availability

No data was used for the research described in the article.

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- Department of Computer Engineering, Prince Sattam bin Abdulaziz University, Alkharj, Saudi Arabia, and
- Faculty of Computing and Information Technology, King Abdulaziz University, Jeddah, Saudi Arabia.

Submission Declaration

It is confirmed that

- The work described has not been published previously,
- It is not under consideration for publication elsewhere,
- The manuscript is approved by all authors.
- If accepted, it will not be published elsewhere in the same form, in English or in any other language, including electronically without the written consent of the copyright-holder.

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